

FLASH DRUM VER 1

Operator Training Simulator · D-400 with E-400 feed preheater

Chulalongkorn University · Department of Chemical Engineering

HAZOP REPORT

Hazard and Operability Study of the D-400 flash drum unit

DOCUMENT NO. : FD-V1-SA-HZP-001
DOCUMENT TITLE : HAZOP REPORT
STANDARD : IEC 61882:2016
DRAWINGS : FD-400-PID-001 · FD-400-PID-002
STUDY TYPE : Teaching study, basic-design level

Teaching document generated from the HAZOP tab of the D-400 simulator. It is not a study of a real plant.

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
A	06-Oct-26	Issued for teaching	A. Soottitantawat			

HAZOP CLOSE-OUT SUMMARY

Project: Flash Drum Ver 1 · D-400 Operator Training Simulator · Document FD-V1-SA-HZP-001 rev A · 31 actions, all open at issue. The action party fills "Action taken" and the HAZOP leader changes the status to CLOSED when the close-out sheet (Appendix 3) is signed.

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1/N2 Feed supply and flow control / E-400 feed preheater, feed (tube) side	R-01	N1.1, N1.5, N2.5, N7.2, N7.3	Interlock the hot oil to the feed: close XV-409 on low-low feed flow (an element independent of FT-400) and whenever XV-408 closes, so E-400 is never heated with the tube side blocked in or stagnant.	Process / Instrument		OPEN	Risk 12 High
N1/N2 Feed supply and flow control / E-400 feed preheater, feed (tube) side	R-02	N1.1, N1.5, N2.5	Fit a thermal relief valve on the E-400 tube side (between XV-408 and FCV-400), sized for blocked-in liquid heated by hot oil at 180 °C, discharging to a closed system.	Process		OPEN	Risk 12 High
N1 Feed supply and flow control	R-03	N1.4	Add a check valve in the feed line downstream of XV-408 so D-400 and E-400 cannot flow back into the supply when the supply pump stops.	Process / Mechanical		OPEN	Risk 6 Medium
N6 Liquid line to the downstream unit	R-04	N6.3	Add a check valve in the bottoms line downstream of HV-411 against back-flow from the downstream receiver (40 kPa g) when D-400 pressure is low.	Process / Mechanical		OPEN	Risk 6 Medium
N5 Vapour line to the header	R-05	N5.4	Add a check valve in the vapour line (or a low-pressure closure of HV-421 by procedure) so header gas cannot flow back into D-400 when the drum falls to the N2 blanket pressure.	Process		OPEN	Risk 9 Medium
N1 Feed supply and flow control	R-06	N1.2, N1.6	Add a feed supply pressure indicator with low alarm upstream of FCV-400 and a FIC-400 output-high alarm, so loss of supply head is seen before FCV-400 runs out of travel.	Instrument		OPEN	Risk 8 Medium
N1 Feed supply and flow control	R-07	N1.3	Limit the feed with FCV-400 wide open (travel stop or restriction orifice), or show that D-400, the mesh pad, PCV-420 and the downstream receiver handle that flow.	Process		OPEN	Risk 9 Medium
N2/N3 E-400 feed preheater, feed (tube) side / Hot-oil side of E-400	R-08	N2.1, N2.3, N3.3	Evaluate by LOPA an independent high-temperature trip on the E-400 outlet (own element) closing XV-409 below the bubble point of the feed at the FCV-400 inlet pressure.	Process / Instrument		OPEN	Risk 9 Medium
N2 E-400 feed preheater, feed (tube) side	R-09	N2.3	Add a second temperature element at the E-400 outlet with a deviation alarm against TT-401.	Instrument		OPEN	Risk 9 Medium
N2/N3/N6 E-400 feed preheater, feed (tube) side / Hot-oil side of E-400 / Liquid line to the downstream unit	R-10	N1.2, N2.7, N3.2, N6.4, N7.6	Run the Rev B cascades (TIC-401 → FIC-405, LIC-400 → FIC-410) as the normal configuration; state the minimum stable feed rate for TIC-401 in the start-up procedure.	Instrument / Operations		OPEN	Risk 8 Medium
N3 Hot-oil side of E-400	R-11	N2.2, N3.1, N3.2, N3.4, N3.5	Add hot-oil supply alarms: TI-402 low and high, FI-405 low; agree with utilities to warn the unit before hot-oil changes.	Instrument / Operations		OPEN	Risk 12 High
N2 E-400 feed preheater, feed (tube) side	R-12	N2.2, N2.4	Add an E-400 performance display (UA from TI-400, TT-401, TI-402, TI-403 and FI-405) and a TCV-401 output-high alarm; plan cleaning on the trend.	Process / Instrument		OPEN	Risk 12 High

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N2/N3 E-400 feed preheater, feed (tube) side / Hot-oil side of E-400	R-13	N2.6, N3.6	Confirm the E-400 tube-rupture case: shell (hot-oil) design pressure against the tube side (two-thirds rule) and relief of the hot-oil system; test the hot oil for light ends (flash point) periodically.	Mechanical / Process		OPEN	Risk 8 Medium
N4 D-400 flash drum	R-14	N2.1, N3.3, N4.3, N4.5, N4.12	Verify PSV-400 (450 kPa g) for every relief case: blocked vapour outlet, external fire, TCV-401 failed to full E-400 and PCV-430 failed open; record them in the relief study.	Process		OPEN	Risk 10 High
N4 D-400 flash drum	R-15	N4.3, N4.8	Fit a restriction orifice after PCV-430 so a failed-open regulator on the 700 kPa g N2 header cannot exceed PSV-400 capacity.	Process		OPEN	Risk 10 High
N4 D-400 flash drum	R-16	N2.2, N3.1, N4.4, N4.9	Confirm the D-400 vacuum rating, or the N2 blanket capacity, for the cold-drum cases (hot oil lost, cold feed, fast depressuring); add a PT-420 low-low alarm.	Process / Mechanical		OPEN	Risk 12 High
N4/N5 D-400 flash drum / Vapour line to the header	R-17	N4.3, N5.1, N5.3	Add a high-pressure pre-alarm on PT-420 between the PIC-420 HI alarm (130 kPa g) and PSHH-426 (250 kPa g), with a written response: restore the vapour path (HV-422), call the header owner.	Instrument / Operations		OPEN	Risk 12 High
N5 Vapour line to the header	R-18	N5.2, N7.8	Check the fuel-gas header (and flare) capacity for PCV-420 failed open and for drum depressuring; consider a travel stop on PCV-420.	Process		OPEN	Risk 9 Medium
N4 D-400 flash drum	R-19	N4.1, N4.10, N4.11, N6.1	Add an independent level transmitter LT-400B with its own high-level alarm, and comparison alarms LT-400 vs the LSHH-416/LSLL-417 level sensors and PT-420 vs PSHH-426; read LG-400 on every round.	Instrument / Operations		OPEN	Risk 12 High
N4/N6 D-400 flash drum / Liquid line to the downstream unit	R-20	N4.1, N4.2, N6.2, N7.5	Review the I-2 action: when XV-418 closes, alarm and cut the feed (or require it by procedure) so the level does not climb to LSHH-416; take the gas blow-by scenario to LOPA.	Process / Instrument		OPEN	Risk 12 High
N6 Liquid line to the downstream unit	R-21	N4.2, N6.2	Confirm the downstream receiver relief for gas blow-by through LCV-400 and HV-413 fully open at the highest drum pressure, and that the LSLL-417 delay (5 s) is short enough.	Process		OPEN	Risk 10 High
N4/N5 D-400 flash drum / Vapour line to the header	R-22	N4.6, N4.7, N5.5, N5.6	Carry-over: inspect the mesh pad at each turnaround, provide an antifoam injection point, slope the vapour line to the drum and confirm the header knock-out drum and drains.	Mechanical / Process		OPEN	Risk 9 Medium
N1/N6 Feed supply and flow control / Liquid line to the downstream unit	R-23	N1.8, N1.9, N4.6, N5.5, N6.5, N6.7	Sample the feed at every tank switch and the products each shift; add a flat-line (frozen signal) alarm on AI-410 and AI-420.	Operations / Instrument		OPEN	Risk 9 Medium
N1 Feed supply and flow control	R-24	N1.10	Draw water from the feed tank before a switch and check the feed for free water; include a D-400 low-point water check in the rounds.	Operations		OPEN	Risk 6 Medium
N4/N7 D-400 flash drum / Procedures	R-25	N6.6, N7.7, N7.8	Write the drain and depressuring procedure against auto-refrigeration: slow, attended, double valves on HV-415, watch TI-410; confirm the minimum design metal temperature of D-400 bottoms, HV-415 and the drain line.	Process / Operations		OPEN	Risk 12 High
N7 Procedures	R-26	N4.9, N7.1	Make an oxygen check at the N2 sweep vent (portable meter) a hold point before hydrocarbon is admitted; sweep all dead legs (LG-400, instrument nozzles, bottoms line).	Operations		OPEN	Risk 10 High

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N7 Procedures	R-27	N7.2, N7.3, N7.4, N7.9	Positive isolation for maintenance (blinds or double block and bleed at the feed, hot-oil, vapour and liquid battery limits), lock-out, depressure and N2 purge before opening; review the start-up and shutdown procedures against this study.	Operations		OPEN	Risk 10 High
N3/N7 Hot-oil side of E-400 / Procedures	R-28	N3.7, N7.10	SIS management: proof tests of LSHH-416, LSLL-417, PSHH-426 and XV-408/409/418 at the interval set by LOPA; bypass permit with time limit; bypass status shown on the DCS.	Instrument / Operations		OPEN	Risk 10 High
N5 Vapour line to the header	R-29	N4.5, N5.7	Provide gas detection at the D-400 valve stations (to be set in the HAZID follow-up).	Safety		OPEN	Risk 10 High
N7 Procedures	R-30	N5.1, N6.1, N7.5	Train operators on the simulator on the scenarios of the dynamic check (loss of the vapour path, LCV-400 failures, TT-401 fault, hot-oil loss).	Operations		OPEN	Risk 12 High
N1/N4 Feed supply and flow control / D-400 flash drum	R-31	N1.11, N4.12	Add a DCS material balance (FT-400 against FI-410 + FI-420) with a deviation alarm, to reveal meter errors and a passing PSV-400.	Instrument		OPEN	Risk 6 Medium

HAZOP REPORT

Hazard and Operability Study of the D-400 flash drum unit
Flash Drum Ver 1 · D-400 Operator Training Simulator

	Role	Name	Signature	Date
Prepared by	HAZOP scribe			
Reviewed by	HAZOP leader (chair)			
Approved by	Plant / project manager			

This report records a teaching HAZOP of the flash drum unit drawn in the simulator. Names, signatures and dates are filled in by the class team that runs the study; the risk matrix is an example for teaching, not a corporate matrix.

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Executive summary

A HAZOP of the D-400 flash drum unit — the feed line with XV-408 and the let-down valve FCV-400, the E-400 feed preheater (feed and hot-oil sides, XV-409 and the 3-way TCV-401), the flash drum D-400 with its level and pressure control, N2 blanket, PSV-400 and SIS switches, the vapour line to the fuel-gas header and the liquid line to the downstream unit, and the start-up, shutdown and maintenance procedures — was carried out with the IEC 61882 guide-word method. The unit was divided into **7 nodes**; **61 deviations** were recorded and **31 recommendations** raised.

Nodes	Deviations	Recommendations	Very high	High	Medium	Low
7	61	31	0	15	45	1

Risk is ranked with the existing safeguards credited (Section 3.4). No deviation was ranked very high; the high-ranked deviations are mostly those where an instrumented trip or the relief valve is the main protection, and they are carried to the LOPA, the SRS of the SIFs (FD-V1-IN-SRS-001) and the cause and effect diagram (FD-V1-IN-CE-001).

Key findings

- **N1 flow no:** FIC-400 LO alarms at 00:02; nothing trips. The true E-400 outlet reaches 123.7 °C with the feed stopped, because hot oil stays in the shell until TIC-401 has moved TCV-401 to bypass. No flow plus heat is the blocked-in case behind R-01 and R-02; the drum only falls to 81 kPa g.
- **N1 flow more:** the feed rises to 28.4 m³/h and the vapour to 1094 Nm³/h (design 611); FIC-400 HI alarms at 00:02, TIC-401 cannot hold (64 °C) and the level peaks at 60 %. The higher flow also takes more pressure drop ahead of FCV-400, so the hot feed is above its bubble point at the valve inlet from 00:05. Nothing trips: the alarms, not a trip, protect the header and the receiver (R-07).
- **N2 temperature more:** the true hot feed reaches 101.9 °C, 12.5 °C above its bubble point at the FCV-400 inlet from 00:49; TIC-401 HI alarms at 00:22, AI-420 falls to 67.8 wt%. Nothing trips: only the alarm stands between the operator and vapour ahead of the let-down valve (R-08).
- **N2 measurement other than:** TIC-401 still reads its set point while the true feed rises to 89.0 °C (margin to the bubble point at worst 0.4 °C); the first alarm is TIC-401 LO at 00:21 (a low-temperature alarm while the feed is really hotter: it points the wrong way). A second element with a deviation alarm (R-09) is the only direct detection.
- **N3 flow no:** the drum pressure falls to 24.7 kPa g, below the 30 kPa g vapour header at 10:59, the level rises to 68 % and AI-410 reaches 33.4 wt%; nothing trips. The N2 blanket does not hold the drum above the header, so header gas can flow back without a check valve (R-05, R-16).
- **N5 flow no:** PIC-420 HI alarms at 01:46 and SIS I-1 trips on PSHH-426 at 16:05; PSV-400 is not reached (pressure peak 250 kPa g). The operator has the time between the alarm and the trip to open HV-422: a pre-alarm and a written response (R-17) make that window usable.
- **N6 flow no:** FIC-410 LO alarms at 00:02, LIC-400 HI at 07:47; SIS I-1 trips on LSHH-416 at 10:26 (level 90 %). The independent LSHH-416 acts before liquid reaches the vapour nozzle; this is the high-level scenario for LOPA.
- **N6 flow more:** the level falls and I-2 closes XV-418 at 09:06, so gas blow-by is stopped; with the feed still running the level then climbs to 86 % in the 30 min (nothing trips). I-2 turns a low-level event into a high-level one unless the operator cuts the feed (R-20).
- **N5 flow more:** the vapour peaks at 3217 Nm³/h, about 5.3 times design, while the drum depressures to 33 kPa g and the liquid cools to 38.4 °C; nothing trips. The fail-open PCV-420 is safe for the drum and a load case for the header (R-18).
- **N4 pressure less:** every valve goes to its fail position: the feed stops, PCV-420 opens and the vapour peaks at 3001 Nm³/h while the drum falls to 30 kPa g; nothing trips. The fail positions leave D-400 safe; the header must take the depressuring load (R-18).

Actions by action party

Action party	Number	Action No.
Process	17	R-01, R-02, R-03, R-04, R-05, R-07, R-08, R-12, R-13, R-14, R-15, R-16, R-18, R-20, R-21, R-22, R-25
Instrument	13	R-01, R-06, R-08, R-09, R-10, R-11, R-12, R-17, R-19, R-20, R-23, R-28, R-31
Operations	11	R-10, R-11, R-17, R-19, R-23, R-24, R-25, R-26, R-27, R-28, R-30
Mechanical	5	R-03, R-04, R-13, R-16, R-22
Safety	1	R-29

An action shared by two parties (e.g. Instrument / Operations) is counted under both.

1 Introduction

1.1 Purpose

The HAZOP identifies how the D-400 unit can deviate from its design intent, what the deviations lead to, whether the existing safeguards are adequate, and what should be changed. It also finds operability problems: deviations that harm no one but stop the unit, put a product off specification or upset the users downstream. The study feeds the safety layers, LOPA and SRS of the SIFs, and the cause and effect diagram FD-V1-IN-CE-001.

1.2 The unit

A liquid n-butane / n-hexane feed (16 m³/h, 35 wt% butane) passes the SIS valve XV-408 and FT-400, is heated to about 74 °C in E-400 by hot oil through the 3-way TCV-401 (XV-409 on the hot-oil supply), and is let down through FCV-400 into the vertical flash drum D-400 at about 103 kPa g. Vapour (about 77 wt% butane) leaves through the mesh pad and PCV-420 to the fuel-gas header; liquid (about 27 wt% butane) leaves through XV-418 and LCV-400 to a downstream unit. An N₂ blanket (PCV-430) holds a minimum pressure, PSV-400 protects the drum, and the SIS trips the feed and hot oil on high level (LSHH-416) or high pressure (PSHH-426) and closes the bottoms on low level (LSLL-417).

1.3 Objectives

- Identify hazards to people, the environment and the plant from deviations of each node.
- Identify operability problems: product specifications, carry-over to the fuel-gas users, upsets of the downstream unit.
- Judge the existing safeguards and rank the risk with the example matrix.
- Raise recommendations and pass the high-ranked scenarios to LOPA.
- Check selected deviations dynamically on the simulator model (Section 5.3).

2 Scope

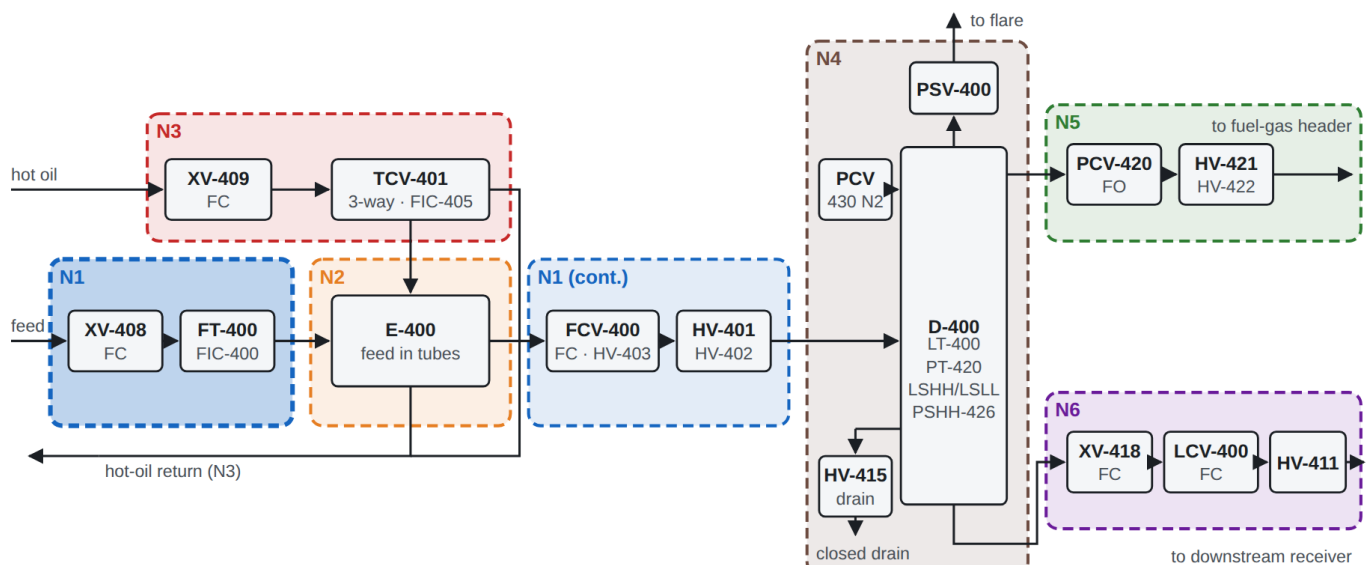
Boundaries: scope, objectives, team and risk matrix. Scope here: from the feed battery limit (upstream of XV-408) through E-400 (both sides, from XV-409), D-400 with PSV-400 and the N₂ blanket, to the vapour battery limit (downstream of HV-421) and the liquid battery limit (downstream of HV-411). Excluded: the supply tankage and pump, the hot-oil heater, the fuel-gas header, the flare and the downstream unit (their interfaces are recorded as causes and consequences).

2.1 Nodes

Node	Description	Design intent	Design conditions	Equipment / instruments
N1	Feed supply and flow control: supply → XV-408 → FT-400 → (E-400) → FCV-400 → HV-401 → D-400	Deliver 16.0 m ³ /h of liquid feed (35 wt% C ₄ , 10 °C) from the supply (650 kPa g shut-off) through E-400 and the let-down valve FCV-400 (fail closed) at the rate FIC-400 sets.	16.0 m ³ /h; supply ≈ 600 kPa g at design flow; FCV-400 inlet ≈ 550 kPa g, OUT 73.5 %; D-400 102.7 kPa g	Feed line from tankage, XV-408 (fail closed, SIS I-1), FT-400 orifice, FCV-400 with block valves and globe bypass HV-403, feed block HV-401 with bypass HV-402 (NC) <i>FT-400/FIC-400, TI-400, XV-408</i>
N2	E-400 feed preheater, feed (tube) side: E-400 inlet → TT-401 → FCV-400 inlet	Heat the feed from 10 to 74 °C (TIC-401 on TT-401) and keep it liquid: well below its bubble point at the FCV-400 inlet (≈ 89 °C at 550 kPa g).	10 → 74 °C, duty ≈ 420 kW; bubble pressure of the hot feed ≈ 359 kPa g vs ≈ 550 kPa g at the FCV-400 inlet	E-400 shell-and-tube (feed in the tubes, hot oil in the shell), outlet line to FCV-400 <i>TT-401/TIC-401, TI-400; TCV-401 acts on the hot-oil side (N3); FIC-405 in Rev B</i>
N3	Hot-oil side of E-400: header → XV-409 → TCV-401 (3-way) → E-400 shell → return	Pass 10.98 t/h of hot oil at 180 °C (header ΔP 150 kPa) through the E-400 shell by the 3-way TCV-401 (fails to full bypass) and return it at ≈ 120 °C; isolate it with XV-409 on SIS I-1.	180 °C in, 120 °C out; 10.98 t/h through E-400 (TCV-401 30.5 % to E-400); PDI-406 150 kPa	Hot-oil supply and return lines, XV-409 (fail closed), TCV-401 3-way valve and bypass, E-400 shell <i>TI-402, PDI-406, FT-405/FIC-405, TI-403, XV-409</i>

Node	Description	Design intent	Design conditions	Equipment / instruments
N4	D-400 flash drum: level, pressure, N2 blanket, PSV-400, SIS switches, mesh pad, drain	Flash the hot feed at 102.7 kPa g and 49.2 °C; separate 611 Nm ³ /h of vapour from the liquid through the mesh pad; hold 50 % level (LIC-400, span 3 m) and 102.7 kPa g (PIC-420); N2 blanket 30 kPa g; relief by PSV-400 at 450 kPa g.	102.7 kPa g, 49.2 °C, level 50 %; MAWP 450 kPa g at 150 °C (P&ID)	D-400 1.6 m ID × 4.8 m T/T with mesh pad; LG-400; PSV-400 to flare; PCV-430/HV-430 N2 from the 700 kPa g header; HV-415 drain to closed drain <i>LT-400/LIC-400, LG-400, PT-420/PIC-420, TI-410, AI-400, LSHH-416, LSL-417, PSHH-426 (own sensors, SIS I-1/I-2)</i>
N5	Vapour line to the header: D-400 top → FE-420 → PCV-420 → HV-421 → fuel-gas header	Send 611 Nm ³ /h of vapour (76.8 wt% C4, spec ≥ 75) through PCV-420 (fail open, OUT 38.7 %) and HV-421 to the fuel-gas header at 30 kPa g, holding D-400 at 102.7 kPa g.	611 Nm ³ /h (1714 kg/h); 102.7 kPa g in the drum → 30 kPa g in the header	Vapour line, FE-420, PCV-420 with block valves and globe bypass HV-423, HV-421 with bypass HV-422 (NC) <i>PT-420/PIC-420, FI-420, AI-420; PSHH-426 on the drum top</i>
N6	Liquid line to the downstream unit: D-400 bottom → FE-410 → XV-418 → LCV-400 → HV-411 → receiver	Send 14.0 m ³ /h of liquid (26.6 wt% C4, spec ≤ 28) through XV-418 (fail closed), LCV-400 (fail closed, OUT 63.4 %) and HV-411 to the downstream receiver at 40 kPa g, driven by drum pressure plus static head.	14.0 m ³ /h (8495 kg/h) at 49.2 °C; 102.7 kPa g plus static head → 40 kPa g	Bottoms line, FE-410, XV-418 (SIS I-2), LCV-400 with block valves and globe bypass HV-413, HV-411 with bypass HV-412 (NC) <i>FT-410/FIC-410 (Rev B), AI-410, TI-410; LSL-417 acts through I-2</i>
N7	Procedures: start-up with N2 sweep, shutdown, depressuring and draining, maintenance	Start-up: sweep D-400 with N2, pressurise on the blanket, establish feed, then hot oil, then line out. Shutdown: hot oil off, feed off, depressure to the header, drain to the closed drain. Maintenance: isolate, depressure, purge, blind.	Procedure steps, targets and time limits in the Exercises tab	Whole unit; SIS reset and permissives; HV-415 drain; PCV-430/HV-430 <i>AI-400; SIS reset permissives (level below 85 %, pressure below 230 kPa g; XV-418 only above 10 % level); procedures in the Exercises tab</i>

2.2 Node map



N7 (not drawn): start-up with N2 sweep, shutdown, depressuring and draining, maintenance

Figure 1. D-400 divided into HAZOP nodes (N7, the procedures node, covers the whole unit).

3 Methodology

HAZOP is a structured, team-based examination of a defined system. Each part of the design is compared with its **design intent**; **guide words** combined with process **parameters** generate **deviations**; for each credible deviation the team records causes, consequences, existing safeguards and, where the risk is not acceptable, recommendations. It finds hazards (harm to people, environment, assets) and operability problems (the unit cannot make on-spec product or cannot be run as intended).

3.1 References

Reference	What it contributes to this study
IEC 61882:2016 <i>Hazard and operability studies (HAZOP studies) – Application guide</i>	Definitions, the four phases, the guide words, team roles, recording
CCPS <i>Guidelines for Hazard Evaluation Procedures</i>	Worksheet format, node selection, consequences judged without safeguards, risk ranking, hand-over to LOPA
IEC 61511 (functional safety, SIS)	How HAZOP scenarios that rely on SIS I-1 and I-2 go to LOPA to set the required SIL of LSHH-416, LSL-417 and PSHH-426 loops
ISO 31000 and the owner's risk matrix	Risk criteria used to decide whether the safeguards are adequate
API 520 / API 521	Relief sizing and the relief cases for PSV-400 (blocked outlet, external fire, control-valve failure, regulator failure), depressuring, flare loads
ASME Boiler and Pressure Vessel Code, Section VIII	D-400 design pressure (MAWP), design temperature, minimum design metal temperature, overpressure protection
Thailand: Department of Industrial Works regulation on hazard identification, risk assessment and risk-management plans (B.E. 2543)	Legal duty to identify hazards (HAZOP is an accepted method) and to prepare a risk-management plan for the factory
OSHA 29 CFR 1910.119 (PSM), where applicable	Process hazard analysis for a covered process (flammable liquids and gases), revalidated at least every five years

3.2 The four phases

- **Definition:** scope, objectives, team and risk matrix. Scope here: from the feed battery limit (upstream of XV-408) through E-400 (both sides, from XV-409), D-400 with PSV-400 and the N2 blanket, to the vapour battery limit (downstream of HV-421) and the liquid battery limit (downstream of HV-411). Excluded: the supply tankage and pump, the hot-oil heater, the fuel-gas header, the flare and the downstream unit (their interfaces are recorded as causes and consequences).
- **Preparation:** collect the P&ID (both sheets), the stream table and design limits, the SIS cause-and-effect chart, the alarm list, the procedures; divide the unit into nodes; write each design intent; choose parameters and guide words.
- **Examination:** for each node → state the design intent → choose a parameter → apply each guide word → is the deviation meaningful? → causes → consequences assuming the safeguards fail → existing safeguards → rank the risk → recommend → next guide word. Every meaningful deviation is recorded, including those with adequate safeguards (**full recording**), so the reasoning can be audited later.
- **Documentation and follow-up:** worksheets, recommendation register with owners and dates, close-out, and revalidation after changes (MOC) and periodically.

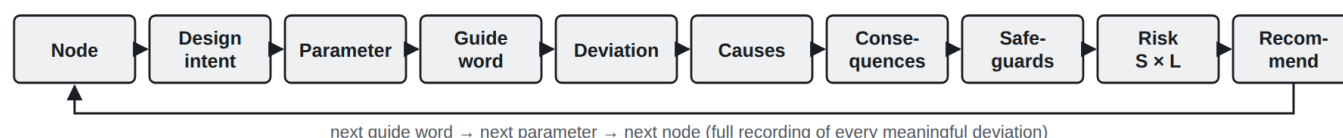


Figure 2. HAZOP examination loop.

Consequences are judged with no credit for safeguards (the worst credible outcome if every safeguard failed); the likelihood L is then judged with the existing safeguards working as designed. Keeping the two apart stops a safeguard being counted twice.

3.3 Guide words and parameters

Guide word	Meaning (IEC 61882)
NO / NOT	Complete negation of the design intent
MORE	Quantitative increase
LESS	Quantitative decrease

Guide word	Meaning (IEC 61882)
AS WELL AS	Qualitative increase: something extra is present
PART OF	Qualitative decrease: only part of the intent is achieved
REVERSE	Logical opposite of the intent (e.g. back-flow)
OTHER THAN	Complete substitution: something else happens
EARLY	Relative to clock time: happens too soon
LATE	Relative to clock time: happens too late
BEFORE	Relative to order or sequence: a step comes too soon
AFTER	Relative to order or sequence: a step comes too late

Parameters used (19): Flow (13), Pressure (8), Temperature (8), Composition (8), Measurement (5), Heat transfer (1), Integrity (2), Control (1), Safety system (2), Level (2), Separation (1), Relief (1), Purge (N2 sweep) (1), Sequence (2), Timing (2), Feed rate (1), Temperature (drain) (1), Depressuring (1), Maintenance (1).

A deviation is recorded only when it is physically meaningful ("REVERSE temperature" is not). Guide words may also be combined with a parameter in plain words ("less level", "no flow", "hot oil as well as feed").

3.4 Risk ranking

S \ L	L1	L2	L3	L4	L5
S5	5	10 (8)	15	20	25
S4	4	8 (8)	12 (6)	16	20
S3	3	6 (4)	9 (13)	12 (1)	15
S2	2	4	6 (10)	8 (10)	10
S1	1	2	3	4 (1)	5

Number in brackets: deviations of this study in the cell. Risk $R = S \times L$.

S	Severity (worst credible consequence, no credit for safeguards)
1	No injury; minor upset; no release
2	First aid; off-spec product; short outage; small contained release
3	Lost-time injury; unit down for days; local release or small fire
4	Single fatality or permanent disability; major equipment damage; fire in the unit
5	Multiple fatalities; vessel rupture or explosion; fire or release beyond the unit

L	Likelihood with the existing safeguards
1	Remote: $< 10^{-4}$ /yr
2	Unlikely: 10^{-4} – 10^{-3} /yr
3	Possible in the plant life: 10^{-3} – 10^{-2} /yr
4	Likely, several times in the plant life: 10^{-2} – 10^{-1} /yr
5	Frequent: $> 10^{-1}$ /yr

1–4 Low acceptable · 5–9 Medium reduce if practicable (ALARP), review the safeguards · 10–16 High recommendation required, take to LOPA · 20–25 Very high not acceptable, act before operation

4 HAZOP sessions

4.1 Schedule

The study is run in class as a teaching HAZOP. A typical plan is three sessions; the team records the dates and attendance on this page.

Session	Nodes	Date	Remark
1	N1 – N3 (feed and let-down, E-400 feed side, hot-oil side)		
2	N4 – N6 (D-400, vapour line, liquid line)		
3	N7 (procedures); review of recommendations		

4.2 Team

Role	Responsibility	Name	Attended sessions
HAZOP leader (chair)	Independent of the design; runs the method, keeps the team on the node, decides when a deviation is covered		
Scribe (recorder)	Writes the worksheet live and reads back each recommendation		
Process engineer (designer)	Design intent, balances, bubble-point margins, relief cases		
Operations representative	How D-400 is really run: start-up with N2 sweep, rounds, draining, tank switches		
Instrument / control engineer	Loops, fail positions (FC/FO), SIS logic, independence of LSHH-416, LSLH-417, PSHH-426		
Safety / risk engineer	Risk matrix, LOPA follow-up, regulations		
Specialists as needed	Mechanical (E-400, vessel ratings), utilities (hot oil, N2, instrument air), downstream and header owners		

4.3 Documents reviewed

Document	Title / content
FD-400-PID-001	D-400 P&ID sheet 1, piping and instruments (Appendix 2)
FD-400-PID-002	D-400 P&ID sheet 2, control loops, loop data and SIS cause and effect (Appendix 2)
FD-400-PFD-001	Process flow diagram with the stream flags
FD-V1-IN-CE-001 rev R0	Cause and effect diagram of the SIS and alarms
FD-V1-IN-SRS-001 rev R0	Safety requirements specification of the SIFs
Simulator tabs	Process, Design of the unit operation, Control loop design, alarm and trip list (Workbook tab), start-up and shutdown procedures (SOP tab)

4.4 Nodes reviewed and guide-word coverage

Node	NO / NOT	MORE	LESS	AS WELL AS	PART OF	REVERSES	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows
N1	1	4	3	1	·	1	1	·	·	·	·	11
N2	·	2	2	·	·	·	3	·	·	·	·	7
N3	1	3	2	·	·	·	1	·	·	·	·	7
N4	·	3	2	2	1	·	4	·	·	·	·	12
N5	1	2	2	·	·	1	1	·	·	·	·	7
N6	1	3	1	·	·	1	1	·	·	·	·	7
N7	·	1	3	·	·	·	2	1	1	1	1	10
All	4	18	15	3	1	3	13	1	1	1	1	61

Numbers: rows recorded. A dot: the guide word was considered and gave no meaningful deviation in that node. EARLY, LATE, BEFORE and AFTER apply mainly to the procedure node N7. The second table checks that every node has flow, pressure and temperature (or their equivalents) covered.

5 Findings and recommendations

5.1 Risk profile

Node	Deviations	High	Medium	Low	Recommendations
N1 Feed supply and flow control	11	0	10	1	R-01, R-02, R-03, R-06, R-07, R-23, R-24, R-31
N2 E-400 feed preheater, feed (tube) side	7	2	5	0	R-01, R-02, R-08, R-09, R-10, R-12, R-13
N3 Hot-oil side of E-400	7	0	7	0	R-08, R-10, R-11, R-13, R-28
N4 D-400 flash drum	12	6	6	0	R-14, R-15, R-16, R-17, R-19, R-20, R-22, R-25, R-31
N5 Vapour line to the header	7	1	6	0	R-05, R-17, R-18, R-22, R-29
N6 Liquid line to the downstream unit	7	2	5	0	R-04, R-10, R-20, R-21, R-23
N7 Procedures	10	4	6	0	R-25, R-26, R-27, R-28, R-30
Total	61	15	45	1	31

High-ranked deviations

Ref	Deviation	Consequence	SxL	Rec.
N2.2	Temperature LESS: Hot feed below set point	Little or no flash: vapour product falls, drum pressure falls to the N2 blanket and can fall below it, level rises; bottoms far above the 28 wt% spec sent downstream.	3x4 = 12	R-11, R-12, R-16
N2.5	Pressure MORE: Tube side blocked in while hot oil flows	Trapped liquid expands and vaporises: tube-side pressure far above design, tube or flange failure, feed into the hot oil or to atmosphere, fire.	4x3 = 12	R-01, R-02
N4.1	Level MORE: High level	Liquid reaches the mesh pad and vapour nozzle: liquid slugs into the vapour line and fuel-gas header (burner flooding and fire at users); vapour space lost, pressure rises.	4x3 = 12	R-19, R-20
N4.2	Level LESS: Low level (liquid seal lost)	Vapour blows through LCV-400 into the downstream receiver built for liquid at 40 kPa g: receiver over-pressure, rupture, release; downstream pumps lose suction.	5x2 = 10	R-20, R-21
N4.3	Pressure MORE: High drum pressure	Drum pressure above MAWP (450 kPa g): rupture and large release of C4/C6, fire or explosion; liquid pushed out faster to the receiver.	5x2 = 10	R-14, R-15, R-17
N4.5	Temperature MORE: External fire at D-400	Liquid boils off, the unwetted shell above the liquid overheats and fails below the PSV set pressure; large fire or vessel rupture.	5x2 = 10	R-14, R-29
N4.9	Composition OTHER THAN: Oxygen inside the drum	Flammable mixture inside D-400: internal explosion on ignition (static, pyrophoric scale).	5x2 = 10	R-26, R-16
N4.10	Measurement OTHER THAN: LT-400 reads wrong	LIC-400 moves LCV-400 on a false level: the true level can reach the vapour nozzle or empty unseen.	4x3 = 12	R-19
N5.1	Flow NO: No vapour to the header	Drum pressure rises at the flash rate (N4 pressure MORE): I-1 trip or PSV-400 lift to flare; liquid richer in C4 (AI-410 off spec) meanwhile.	4x3 = 12	R-17, R-30
N6.1	Flow NO: No liquid outflow	D-400 level rises (N4 level MORE) to the high-high switch within minutes (see dynamic check); downstream unit starved.	4x3 = 12	R-19, R-30
N6.2	Flow MORE: More liquid than made; gas blow-by	The drum drains; once the level is lost, vapour at drum pressure blows through into the receiver built for liquid at 40 kPa g: over-pressure, release.	5x2 = 10	R-20, R-21
N7.1	Purge (N2 sweep) LESS: N2 sweep incomplete	Air left in D-400 when hydrocarbon enters: flammable mixture, internal explosion on ignition.	5x2 = 10	R-26
N7.7	Temperature (drain) LESS: Auto-refrigeration while draining	Valve and drain line chilled toward the boiling point of butane (about -0.5 °C at atmospheric pressure): ice on the valve can jam HV-415 open and release C4; cold burns. The model does not reach the carbon-steel minimum design metal temperature (typical -29 °C).	4x3 = 12	R-25
N7.9	Maintenance OTHER THAN: Isolation incomplete	Hydrocarbon or hot oil released when equipment is opened: fire, burns, exposure.	5x2 = 10	R-27

Ref	Deviation	Consequence	SxL	Rec.
N7.10	Safety system OTHER THAN: SIS bypassed or reset with the cause present	High level or high pressure unprotected.	5x2 = 10	R-28

5.2 Recommendations

No.	Recommendation	Node	Raised from	Action by
R-01	Interlock the hot oil to the feed: close XV-409 on low-low feed flow (an element independent of FT-400) and whenever XV-408 closes, so E-400 is never heated with the tube side blocked in or stagnant.	N1/N2	N1.1, N1.5, N2.5, N7.2, N7.3	Process / Instrument
R-02	Fit a thermal relief valve on the E-400 tube side (between XV-408 and FCV-400), sized for blocked-in liquid heated by hot oil at 180 °C, discharging to a closed system.	N1/N2	N1.1, N1.5, N2.5	Process
R-03	Add a check valve in the feed line downstream of XV-408 so D-400 and E-400 cannot flow back into the supply when the supply pump stops.	N1	N1.4	Process / Mechanical
R-04	Add a check valve in the bottoms line downstream of HV-411 against back-flow from the downstream receiver (40 kPa g) when D-400 pressure is low.	N6	N6.3	Process / Mechanical
R-05	Add a check valve in the vapour line (or a low-pressure closure of HV-421 by procedure) so header gas cannot flow back into D-400 when the drum falls to the N2 blanket pressure.	N5	N5.4	Process
R-06	Add a feed supply pressure indicator with low alarm upstream of FCV-400 and a FIC-400 output-high alarm, so loss of supply head is seen before FCV-400 runs out of travel.	N1	N1.2, N1.6	Instrument
R-07	Limit the feed with FCV-400 wide open (travel stop or restriction orifice), or show that D-400, the mesh pad, PCV-420 and the downstream receiver handle that flow.	N1	N1.3	Process
R-08	Evaluate by LOPA an independent high-temperature trip on the E-400 outlet (own element) closing XV-409 below the bubble point of the feed at the FCV-400 inlet pressure.	N2/N3	N2.1, N2.3, N3.3	Process / Instrument
R-09	Add a second temperature element at the E-400 outlet with a deviation alarm against TT-401.	N2	N2.3	Instrument
R-10	Run the Rev B cascades (TIC-401 → FIC-405, LIC-400 → FIC-410) as the normal configuration; state the minimum stable feed rate for TIC-401 in the start-up procedure.	N2/N3/N6	N1.2, N2.7, N3.2, N6.4, N7.6	Instrument / Operations
R-11	Add hot-oil supply alarms: TI-402 low and high, FI-405 low; agree with utilities to warn the unit before hot-oil changes.	N3	N2.2, N3.1, N3.2, N3.4, N3.5	Instrument / Operations
R-12	Add an E-400 performance display (UA from TI-400, TT-401, TI-402, TI-403 and FI-405) and a TCV-401 output-high alarm; plan cleaning on the trend.	N2	N2.2, N2.4	Process / Instrument
R-13	Confirm the E-400 tube-rupture case: shell (hot-oil) design pressure against the tube side (two-thirds rule) and relief of the hot-oil system; test the hot oil for light ends (flash point) periodically.	N2/N3	N2.6, N3.6	Mechanical / Process
R-14	Verify PSV-400 (450 kPa g) for every relief case: blocked vapour outlet, external fire, TCV-401 failed to full E-400 and PCV-430 failed open; record them in the relief study.	N4	N2.1, N3.3, N4.3, N4.5, N4.12	Process
R-15	Fit a restriction orifice after PCV-430 so a failed-open regulator on the 700 kPa g N2 header cannot exceed PSV-400 capacity.	N4	N4.3, N4.8	Process
R-16	Confirm the D-400 vacuum rating, or the N2 blanket capacity, for the cold-drum cases (hot oil lost, cold feed, fast depressuring); add a PT-420 low-low alarm.	N4	N2.2, N3.1, N4.4, N4.9	Process / Mechanical
R-17	Add a high-pressure pre-alarm on PT-420 between the PIC-420 HI alarm (130 kPa g) and PSHH-426 (250 kPa g), with a written response: restore the vapour path (HV-422), call the header owner.	N4/N5	N4.3, N5.1, N5.3	Instrument / Operations
R-18	Check the fuel-gas header (and flare) capacity for PCV-420 failed open and for drum depressuring; consider a travel stop on PCV-420.	N5	N5.2, N7.8	Process
R-19	Add an independent level transmitter LT-400B with its own high-level alarm, and comparison alarms LT-400 vs the LSHH-416/LSLL-417 level sensors and PT-420 vs PSHH-426; read LG-400 on every round.	N4	N4.1, N4.10, N4.11, N6.1	Instrument / Operations
R-20	Review the I-2 action: when XV-418 closes, alarm and cut the feed (or require it by procedure) so the level does not climb to LSHH-416; take the gas blow-by scenario to LOPA.	N4/N6	N4.1, N4.2, N6.2, N7.5	Process / Instrument

No.	Recommendation	Node	Raised from	Action by
R-21	Confirm the downstream receiver relief for gas blow-by through LCV-400 and HV-413 fully open at the highest drum pressure, and that the LSL-417 delay (5 s) is short enough.	N6	N4.2, N6.2	Process
R-22	Carry-over: inspect the mesh pad at each turnaround, provide an antifoam injection point, slope the vapour line to the drum and confirm the header knock-out drum and drains.	N4/N5	N4.6, N4.7, N5.5, N5.6	Mechanical / Process
R-23	Sample the feed at every tank switch and the products each shift; add a flat-line (frozen signal) alarm on AI-410 and AI-420.	N1/N6	N1.8, N1.9, N4.6, N5.5, N6.5, N6.7	Operations / Instrument
R-24	Draw water from the feed tank before a switch and check the feed for free water; include a D-400 low-point water check in the rounds.	N1	N1.10	Operations
R-25	Write the drain and depressuring procedure against auto-refrigeration: slow, attended, double valves on HV-415, watch TI-410; confirm the minimum design metal temperature of D-400 bottoms, HV-415 and the drain line.	N4/N7	N6.6, N7.7, N7.8	Process / Operations
R-26	Make an oxygen check at the N2 sweep vent (portable meter) a hold point before hydrocarbon is admitted; sweep all dead legs (LG-400, instrument nozzles, bottoms line).	N7	N4.9, N7.1	Operations
R-27	Positive isolation for maintenance (blinds or double block and bleed at the feed, hot-oil, vapour and liquid battery limits), lock-out, depressure and N2 purge before opening; review the start-up and shutdown procedures against this study.	N7	N7.2, N7.3, N7.4, N7.9	Operations
R-28	SIS management: proof tests of LSHH-416, LSL-417, PSHH-426 and XV-408/409/418 at the interval set by LOPA; bypass permit with time limit; bypass status shown on the DCS.	N3/N7	N3.7, N7.10	Instrument / Operations
R-29	Provide gas detection at the D-400 valve stations (to be set in the HAZID follow-up).	N5	N4.5, N5.7	Safety
R-30	Train operators on the simulator on the scenarios of the dynamic check (loss of the vapour path, LCV-400 failures, TT-401 fault, hot-oil loss).	N7	N5.1, N6.1, N7.5	Operations
R-31	Add a DCS material balance (FT-400 against FI-410 + FI-420) with a deviation alarm, to reveal meter errors and a passing PSV-400.	N1/N4	N1.11, N4.12	Instrument

5.3 Dynamic check of selected deviations on the simulator

A conventional HAZOP judges consequences by experience. Here selected deviations are run on a copy of the D-400 model, starting from the lined-out design state, with the existing safeguards active (alarms from the alarm table evaluated every second, SIS I-1 and I-2 armed, PSV-400, controllers in their normal modes) and **no operator action**, for up to 30 min or until SIS I-1 trips. The live simulator is not touched. The runs show whether the safeguards act before harm and how much time the operator has.

Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	SIS first-out (mm:ss) and other events	Range of key values (to the trip or 30 min)
N1	Flow NO	HV-401 feed block valve fails closed	FIC-400 LO 00:02 TIC-401 HI 00:42 FI-420 LO 00:52 FIC-410 LO 02:46	no I-1 trip	FIC-400 feed: 0.0 to 16.0 m ³ /h E-400 outlet, true: 74.0 to 123.7 °C D-400 level, true (LSHH/LSLL sensors): 24.6 to 50.0 % D-400 pressure, true (PSHH-426): 80.8 to 102.7 kPa g
N1	Flow MORE	FCV-400 fails wide open (FIC-400 forced to MAN, OUT 100 %)	FIC-400 HI 00:02 TIC-401 LO 00:21 FIC-410 HI 01:46	no I-1 trip hot feed passes its bubble point at the FCV-400 inlet at 00:05	FIC-400 feed: 16.0 to 28.4 m ³ /h D-400 level, true (LSHH/LSLL sensors): 50.0 to 60.4 % FI-420 vapour: 611 to 1094 Nm ³ /h TIC-401 (TT-401 reading): 64.0 to 74.0 °C
N2	Temperature MORE	TCV-401 sends all hot oil to E-400 (TIC-401 forced to MAN, OUT 100 %)	TIC-401 HI 00:22 AI-420 LO 01:14	no I-1 trip hot feed passes its bubble point at the FCV-400 inlet at 00:49	E-400 outlet, true: 74.0 to 101.9 °C Hot feed minus its bubble point at the FCV-400 inlet: -15.4 to 12.5 °C AI-420 vapour C4: 67.8 to 76.8 wt% D-400 pressure, true (PSHH-426): 102.7 to 109.5 kPa g
N2	Measurement OTHER THAN	TT-401 reads 15 °C low (1 min ramp)	TIC-401 LO 00:21 AI-420 LO 02:35	no I-1 trip	TIC-401 (TT-401 reading): 62.7 to 74.0 °C E-400 outlet, true: 74.0 to 89.0 °C Hot feed minus its bubble point at the FCV-400 inlet: -15.4 to -0.4 °C AI-420 vapour C4: 72.2 to 76.8 wt%

Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	SIS first-out (mm:ss) and other events	Range of key values (to the trip or 30 min)
N3	Flow NO	Hot-oil circulation fails	PDI-406 LO 00:01 TIC-401 LO 00:15 FI-420 LO 01:11 PIC-420 LO 03:55 +3 more	no I-1 trip drum pressure below the vapour header (30 kPa g) at 10:59	TIC-401 (TT-401 reading): 10.0 to 74.0 °C D-400 pressure, true (PSHH-426): 24.7 to 102.7 kPa g D-400 level, true (LSHH/LSLL sensors): 50.0 to 68.3 % AI-410 liquid C4: 26.6 to 33.4 wt%
N5	Flow NO	HV-421 vapour block valve fails closed	FI-420 LO 00:01 PIC-420 HI 01:46 FIC-410 HI 03:25 AI-410 HI 03:41 +1 more	I-1: PSHH-426 D-400 pressure high-high at 16:05	D-400 pressure, true (PSHH-426): 102.7 to 250.3 kPa g FI-420 vapour: 0 to 611 Nm ³ /h AI-410 liquid C4: 26.6 to 31.9 wt%
N6	Flow NO	LCV-400 fails closed	FIC-410 LO 00:02 LIC-400 HI 07:47	I-1: LSHH-416 D-400 level high-high (liquid to the vapour line) at 10:26	D-400 level, true (LSHH/LSLL sensors): 50.0 to 90.3 % FIC-410 liquid: 0.0 to 14.0 m ³ /h D-400 pressure, true (PSHH-426): 102.7 to 103.1 kPa g
N6	Flow MORE	LCV-400 fails wide open (LIC-400 forced to MAN, OUT 100 %)	FIC-410 HI 00:02 LIC-400 LO 05:57 FIC-410 LO 09:08 LIC-400 HI 28:35	no I-1 trip LSLL-417 → I-2 closes XV-418 at 09:06	D-400 level, true (LSHH/LSLL sensors): 4.6 to 85.5 % FIC-410 liquid: 0.0 to 32.9 m ³ /h FI-420 vapour: 578 to 638 Nm ³ /h
N5	Flow MORE	PCV-420 fails wide open (PIC-420 forced to MAN, OUT 100 %)	PIC-420 LO 00:07 FIC-410 LO 01:13 TI-410 LO 03:10	no I-1 trip	D-400 pressure, true (PSHH-426): 33.2 to 102.7 kPa g FI-420 vapour: 611 to 3217 Nm ³ /h TI-410 flash temperature: 38.4 to 49.2 °C
N4	Pressure LESS	Instrument air failure (all control valves to their fail positions)	FIC-410 LO 00:02 FIC-400 LO 00:03 PIC-420 LO 00:06 TIC-401 HI 00:45 +2 more	no I-1 trip	FIC-400 feed: 0.0 to 16.0 m ³ /h D-400 pressure, true (PSHH-426): 29.9 to 102.7 kPa g FI-420 vapour: 0 to 3001 Nm ³ /h TI-410 flash temperature: 42.7 to 49.2 °C

What the runs tell the HAZOP team

- **N1 flow no:** FIC-400 LO alarms at 00:02; nothing trips. The true E-400 outlet reaches 123.7 °C with the feed stopped, because hot oil stays in the shell until TIC-401 has moved TCV-401 to bypass. No flow plus heat is the blocked-in case behind R-01 and R-02; the drum only falls to 81 kPa g.
- **N1 flow more:** the feed rises to 28.4 m³/h and the vapour to 1094 Nm³/h (design 611); FIC-400 HI alarms at 00:02, TIC-401 cannot hold (64 °C) and the level peaks at 60 %. The higher flow also takes more pressure drop ahead of FCV-400, so the hot feed is above its bubble point at the valve inlet from 00:05. Nothing trips: the alarms, not a trip, protect the header and the receiver (R-07).
- **N2 temperature more:** the true hot feed reaches 101.9 °C, 12.5 °C above its bubble point at the FCV-400 inlet from 00:49; TIC-401 HI alarms at 00:22, AI-420 falls to 67.8 wt%. Nothing trips: only the alarm stands between the operator and vapour ahead of the let-down valve (R-08).
- **N2 measurement other than:** TIC-401 still reads its set point while the true feed rises to 89.0 °C (margin to the bubble point at worst 0.4 °C); the first alarm is TIC-401 LO at 00:21 (a low-temperature alarm while the feed is really hotter: it points the wrong way). A second element with a deviation alarm (R-09) is the only direct detection.
- **N3 flow no:** the drum pressure falls to 24.7 kPa g, below the 30 kPa g vapour header at 10:59, the level rises to 68 % and AI-410 reaches 33.4 wt%; nothing trips. The N2 blanket does not hold the drum above the header, so header gas can flow back without a check valve (R-05, R-16).
- **N5 flow no:** PIC-420 HI alarms at 01:46 and SIS I-1 trips on PSHH-426 at 16:05; PSV-400 is not reached (pressure peak 250 kPa g). The operator has the time between the alarm and the trip to open HV-422: a pre-alarm and a written response (R-17) make that window usable.
- **N6 flow no:** FIC-410 LO alarms at 00:02, LIC-400 HI at 07:47; SIS I-1 trips on LSHH-416 at 10:26 (level 90 %). The independent LSHH-416 acts before liquid reaches the vapour nozzle; this is the high-level scenario for LOPA.
- **N6 flow more:** the level falls and I-2 closes XV-418 at 09:06, so gas blow-by is stopped; with the feed still running the level then climbs to 86 % in the 30 min (nothing trips). I-2 turns a low-level event into a high-level one unless the operator cuts the feed (R-20).

- **N5 flow more:** the vapour peaks at 3217 Nm³/h, about 5.3 times design, while the drum depressures to 33 kPa g and the liquid cools to 38.4 °C; nothing trips. The fail-open PCV-420 is safe for the drum and a load case for the header (R-18).
- **N4 pressure less:** every valve goes to its fail position: the feed stops, PCV-420 opens and the vapour peaks at 3001 Nm³/h while the drum falls to 30 kPa g; nothing trips. The fail positions leave D-400 safe; the header must take the depressuring load (R-18).

5.4 Follow-up and close-out

- Each recommendation gets an owner and a due date; the owner answers *accepted and done*, *accepted with an alternative* (with justification) or *rejected* (with a justification agreed by the HAZOP leader and the plant manager). The register in part 8 is the tracking list.
- High and very-high rows whose protection relies on SIS I-1 or I-2 (high level, gas blow-by, high pressure) go to **LOPA** to confirm the required SIL (IEC 61511); see the LOPA tab.
- Relief cases raised here (blocked outlet, external fire, TCV-401 failed to E-400, PCV-430 failed open) are closed in the relief study to API 520/521.
- Any change (putting the Rev B cascades in service, new feeds, a changed alarm limit) goes through **management of change**, which asks whether the HAZOP must be updated.
- Revalidate the study periodically (typically every five years) and after incidents or significant changes.
- Use the simulator to train operators on the scenarios found here and to check that alarm responses and procedures work.

Appendix 1 · HAZOP worksheets

Node N1 · Feed supply and flow control

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Node	N1 · Feed supply and flow control: supply → XV-408 → FT-400 → (E-400) → FCV-400 → HV-401 → D-400
Design intent	Deliver 16.0 m ³ /h of liquid feed (35 wt% C4, 10 °C) from the supply (650 kPa g shut-off) through E-400 and the let-down valve FCV-400 (fail closed) at the rate FIC-400 sets.	Design conditions	16.0 m ³ /h; supply ≈ 600 kPa g at design flow; FCV-400 inlet ≈ 550 kPa g, OUT 73.5 %; D-400 102.7 kPa g
Equipment	Feed line from tankage, XV-408 (fail closed, SIS I-1), FT-400 orifice, FCV-400 with block valves and globe bypass HV-403, feed block HV-401 with bypass HV-402 (NC)	Instruments	FT-400/FIC-400, TI-400, XV-408
Drawing	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Document	FD-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	NO	No feed to D-400	- HV-401 feed block valve gate drops (fails closed) - FCV-400 shut (FIC-400 fault, positioner or instrument-air failure: fail closed) - XV-408 closes spuriously - supply pump trips upstream	Production stops. E-400 tubes hold stagnant feed while hot oil still flows: the feed heats toward the hot-oil temperature, vaporises and, if blocked in, over-pressures the tube side. D-400 level and pressure fall toward the N2 blanket.	- FIC-400 LO alarm 5 m³/h - TIC-401 HI alarm 79 °C and TIC-401 moves TCV-401 to bypass - LIC-400 closes LCV-400 - PCV-430 N2 blanket 30 kPa g	3	3	9 M	R-01, R-02	Process · Process / Instrument
N1.2	Flow	LESS	Less feed than set	- Supply pressure falls (upstream pump loses head) until FCV-400 runs out of travel - FT-400 reads high (drift, plugged impulse line) - FCV-400 stiction after packing work - HV-401 partly closed	Lower throughput. At low flow the E-400 outlet runs hotter for the same oil, TIC-401 gain and dead time grow and the loop cycles; flow swings reach the drum as vapour-rate swings.	- FIC-400 LO alarm 5 m³/h - FIC-400 opens FCV-400 - TIC-401 HI/LO alarms	2	4	8 M	R-06, R-10	Instrument · Operations
N1.3	Flow	MORE	More feed than design	- FCV-400 fails wide open (positioner fault, FIC-400 in MAN at a high output) - FT-400 reads low - set point entered too high - globe bypass HV-403 opened with FCV-400 in service	E-400 cannot heat the extra feed (hot feed colder, less vapour per tonne) while the total vapour and liquid rates rise well above design (see dynamic check): level rises, mesh-pad overload and carry-over, header and downstream receiver overloaded.	- FIC-400 HI alarm 20 m³/h - LIC-400 opens LCV-400 - LIC-400 HI alarm 80 % - FI-420 HI alarm 900 Nm³/h - LSHH-416 I-1 at 90 %	3	3	9 M	R-07	Process
N1.4	Flow	REVERSE	Back-flow from E-400 and D-400 toward the supply	- Supply pump stops with D-400 at pressure and XV-408 open - supply pressure below drum pressure during start-up	Hot, light hydrocarbon flows back into the supply line and tank; flashing and hammer in the supply line; D-400 inventory lost.	- FIC-400 LO alarm (FIC-400 then opens FCV-400 further, which helps the back-flow) - XV-408 closes only on SIS I-1	3	2	6 M	R-03	Process / Mechanical
N1.5	Pressure	MORE	Feed line or E-400 tubes over-pressured	- FCV-400, HV-403 and HV-401 closed with the supply pump running (shut-off 650 kPa g) - liquid blocked in between XV-408 and FCV-400 and heated by hot oil	Shut-off head alone is within the line rating, but blocked-in liquid heated by hot oil rises in pressure without limit: flange leak or tube rupture, release and fire.	- No relief on the E-400 tube side on the P&ID - TIC-401 cuts TCV-401 when TT-401 sees the heat (not when flow is zero at the element)	4	2	8 M	R-01, R-02	Process · Process / Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.6	Pressure	LESS	Low supply pressure	- Upstream pump losing head - low supply-tank level - plugged strainer	FCV-400 opens to hold the flow and loses its margin unseen; lower pressure in E-400 lowers the bubble point of the hot feed (vapour ahead of FCV-400).	- FIC-400 LO alarm (only after FCV-400 saturates) - TIC-401	2	3	6 M	R-06	Instrument
N1.7	Temperature	MORE	Warm feed (TI-400 high)	- Feed from a sun-heated tank - upstream cooler out of service	E-400 needs less duty (TCV-401 closes); feed nearer its bubble point in the supply line: upstream pump cavitation.	- TI-400 indication - TIC-401 reduces the hot oil	1	4	4 L	—	—
N1.8	Composition	MORE	Feed lighter (more C4)	Tank switch to a lighter feed (butane toward 45 wt%)	More vapour and higher drum pressure demand on PCV-420; liquid product C4 above the 28 wt% spec sent downstream.	- AI-410 HI alarm 28 wt% - FI-420 HI alarm 900 Nm³/h - PIC-420	2	4	8 M	R-23	Operations / Instrument
N1.9	Composition	LESS	Feed heavier (less C4)	Tank switch to a heavier feed (butane toward 25 wt%)	Less vapour, more liquid; vapour carries more C6 so AI-420 falls toward its 75 wt% spec; LCV-400 opens further.	- AI-420 LO alarm 75 wt% - FI-420 LO alarm 300 Nm³/h - LIC-400	2	4	8 M	R-23	Operations / Instrument
N1.10	Composition	AS WELL AS	Free water in the feed	Water carried from tank bottoms after a tank switch	Water boils in E-400 and D-400 (pressure surges), collects in the drum bottom and goes to the downstream unit; ice or hydrate in the let-down valve at low temperature; corrosion.	- None specific - HV-415 drain	2	3	6 M	R-24	Operations
N1.11	Measurement	OTHER THAN	FT-400 reads wrong	- Orifice plate damaged - impulse line leak or plugged - wrong range after calibration	FIC-400 holds the wrong flow; the true feed rate is unknown; balances do not close and faults are misread.	- FI-410 + FI-420 as a rough balance - LIC-400 trend	2	3	6 M	R-31	Instrument

Node N2 · E-400 feed preheater, feed (tube) side

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Node	N2 · E-400 feed preheater, feed (tube) side: E-400 inlet → TT-401 → FCV-400 inlet
Design intent	Heat the feed from 10 to 74 °C (TIC-401 on TT-401) and keep it liquid: well below its bubble point at the FCV-400 inlet (≈ 89 °C at 550 kPa g).	Design conditions	10 → 74 °C, duty ≈ 420 kW; bubble pressure of the hot feed ≈ 359 kPa g vs ≈ 550 kPa g at the FCV-400 inlet
Equipment	E-400 shell-and-tube (feed in the tubes, hot oil in the shell), outlet line to FCV-400	Instruments	TT-401/TIC-401, TI-400; TCV-401 acts on the hot-oil side (N3); FIC-405 in Rev B
Drawing	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Document	FD-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Temperature	MORE	Hot feed above set point	- TCV-401 driven to full E-400 (TIC-401 in MAN at a high output, positioner fault) - set point entered too high - FIC-405 fault in Rev B	Hot feed approaches or passes its bubble point at the FCV-400 inlet: vapour forms in E-400 and ahead of FCV-400 (two-phase let-down, vibration, flow and level swings); more and heavier vapour (AI-420 off spec); PCV-420 and header overloaded; drum pressure rises.	- TIC-401 HI alarm 79 °C - TI-410 HI alarm 60 °C - AI-420 LO alarm - PIC-420 opens PCV-420 - PSHH-426 I-1 at 250 kPa g - PSV-400	3	3	9 M	R-08, R-14	Process · Process / Instrument
N2.2	Temperature	LESS	Hot feed below set point	- Hot-oil failure - hot-oil supply cooler - E-400 fouling - TCV-401 fails to full bypass (air or signal loss) - feed above E-400 capacity	Little or no flash: vapour product falls, drum pressure falls to the N2 blanket and can fall below it, level rises; bottoms far above the 28 wt% spec sent downstream.	- TIC-401 LO alarm 69 °C - TI-410 LO alarm 40 °C - PIC-420 LO alarm 70 kPa g - AI-410 HI alarm - PCV-430 N2 blanket 30 kPa g	3	4	12 H	R-11, R-12, R-16	Instrument / Operations · Process / Instrument · Process / Mechanical
N2.3	Measurement	OTHER THAN	TT-401 reads low	- Transmitter drift or fault - deposits on the thermowell - wrong calibration	TIC-401 over-heats the feed while the display shows the set point: the true hot-feed temperature rises toward its bubble point at the FCV-400 inlet (see dynamic check); more, heavier vapour; lighter liquid.	- TI-410 HI alarm 60 °C - AI-420 LO alarm 75 wt% - TI-403 rises (indirect) - no second element on the E-400 outlet	3	3	9 M	R-09, R-08	Instrument · Process / Instrument
N2.4	Heat transfer	LESS	E-400 fouling	Deposits from the feed (tank sludge) or hot-oil breakdown products on the shell side	TCV-401 opens slowly to 100 %, then TIC-401 falls below set point (as temperature LESS); more hot oil used.	- TIC-401 LO alarm 69 °C - TI-403 trend	2	4	8 M	R-12	Process / Instrument
N2.5	Pressure	MORE	Tube side blocked in while hot oil flows	- Feed stopped by FCV-400 or HV-401 (not by I-1) with XV-409 open or passing - shutdown in the wrong order	Trapped liquid expands and vaporises: tube-side pressure far above design, tube or flange failure, feed into the hot oil or to atmosphere, fire.	- SIS I-1 closes XV-409 with XV-408 (not on FCV-400 or HV-401 closure) - TIC-401 moves TCV-401 to bypass when TT-401 rises - no tube-side thermal relief on the P&ID	4	3	12 H	R-01, R-02	Process · Process / Instrument
N2.6	Integrity	OTHER THAN	E-400 tube leak	- Tube corrosion, erosion or vibration - thermal shock (cold feed onto hot tubes at start-up)	Feed into the hot oil (feed side ≈ 550 kPa g in normal operation): C4 flashes in the hot-oil return and expansion vessel of other users, fire. Hot oil into the feed (when the feed side is depressured): heavy oil in the D-400 bottoms, off-spec liquid, fouling downstream.	- None specific - AI-410 - hot-oil system relief (off-plot, not on the P&ID)	4	2	8 M	R-13	Mechanical / Process

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.7	Control	OTHER THAN	Hot-feed temperature cycling	• Feed rate near the TIC-401 cycling limit • hot-oil header pressure swings in Rev A • poor tuning	Hot-feed swings move the vapour rate and drum pressure; at the peaks, vapour forms ahead of FCV-400; product quality swings.	• TIC-401 HI/LO alarms • Rev B cascade TIC-401 → FIC-405 • tuning records (Tuning lab)	2	4	8 M	R-10	Instrument / Operations

Node N3 · Hot-oil side of E-400

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Node	N3 · Hot-oil side of E-400: header → XV-409 → TCV-401 (3-way) → E-400 shell → return
Design intent	Pass 10.98 t/h of hot oil at 180 °C (header ΔP 150 kPa) through the E-400 shell by the 3-way TCV-401 (fails to full bypass) and return it at ≈ 120 °C; isolate it with XV-409 on SIS I-1.	Design conditions	180 °C in, 120 °C out; 10.98 t/h through E-400 (TCV-401 30.5 % to E-400); PDI-406 150 kPa
Equipment	Hot-oil supply and return lines, XV-409 (fail closed), TCV-401 3-way valve and bypass, E-400 shell	Instruments	TI-402, PDI-406, FT-405/FIC-405, TI-403, XV-409
Drawing	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Document	FD-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Flow	NO	No hot oil to E-400	- Hot-oil circulation pump trips off-site - XV-409 closes spuriously - TCV-401 fails to full bypass (instrument air or signal loss)	Feed reaches D-400 near its supply temperature: no flash, vapour product stops, drum pressure falls to the N2 blanket and below, level rises, bottoms far off spec.	- PDI-406 LO alarm 100 kPa (header failure only) - TIC-401 LO alarm - TI-410 LO alarm - PIC-420 LO alarm - PCV-430 N2 blanket	3	3	9 M	R-11, R-16	Instrument / Operations · Process / Mechanical
N3.2	Flow	LESS	Less hot oil than demanded	- Header ΔP falls or swings (other users) - TCV-401 sticking - FT-405 reads high (Rev B)	TCV-401 opens further; once it saturates TIC-401 falls (temperature LESS); header swings cycle TIC-401 in Rev A.	- PDI-406 LO alarm 100 kPa - TIC-401 LO alarm - Rev B FIC-405 rejects header swings	2	4	8 M	R-10, R-11	Instrument / Operations
N3.3	Flow	MORE	More hot oil than needed	- TCV-401 fails to full E-400 (positioner) or TIC-401 in MAN at a high output - header ΔP high	Hot feed over-heated: vapour ahead of FCV-400 (see N2 temperature MORE); PSV-400 relief case "TCV-401 failed open".	- PDI-406 HI alarm 200 kPa - TIC-401 HI alarm 79 °C - PSHH-426 - PSV-400	3	3	9 M	R-08, R-14	Process · Process / Instrument
N3.4	Temperature	MORE	Hot oil hotter than 180 °C	Hot-oil heater control fault off-site	At low feed the outlet can approach the oil temperature, far above the bubble point; film boiling and faster fouling on the tube wall.	- TI-402 indication only (no alarm) - TIC-401 closes TCV-401	2	3	6 M	R-11	Instrument / Operations
N3.5	Temperature	LESS	Hot-oil supply cooler	Hot-oil heater cuts back or trips	TCV-401 opens and FI-405 rises to hold TIC-401, leaving little valve margin; a further upset leaves TIC-401 below set point.	- TIC-401 LO alarm 69 °C - TI-402 indication (no alarm)	2	4	8 M	R-11	Instrument / Operations
N3.6	Pressure	MORE	Hot-oil side over-pressured	- E-400 tube rupture lets feed (≈ 550 kPa g) into the shell - return blocked with the supply open	Shell and hot-oil return over-pressured; C4 carried into the hot-oil system (fire at the hot-oil heater or expansion vessel).	- Hot-oil system relief (off-plot, not on the P&ID) - PDI-406 HI alarm	4	2	8 M	R-13	Mechanical / Process
N3.7	Safety system	OTHER THAN	XV-409 fails to close on demand	Valve stuck, solenoid fault, not proof-tested	On an I-1 trip hot oil keeps heating the blocked-in feed; the pressure rise continues.	- XV-408 closes the feed - TIC-401 moves TCV-401 to bypass - PSV-400	4	2	8 M	R-28	Instrument / Operations

Node N4 · D-400 flash drum

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Node	N4 · D-400 flash drum: level, pressure, N2 blanket, PSV-400, SIS switches, mesh pad, drain
Design intent	Flash the hot feed at 102.7 kPa g and 49.2 °C; separate 611 Nm ³ /h of vapour from the liquid through the mesh pad; hold 50 % level (LIC-400, span 3 m) and 102.7 kPa g (PIC-420); N2 blanket 30 kPa g; relief by PSV-400 at 450 kPa g.	Design conditions	102.7 kPa g, 49.2 °C, level 50 %; MAWP 450 kPa g at 150 °C (P&ID)
Equipment	D-400 1.6 m ID × 4.8 m T/T with mesh pad; LG-400; PSV-400 to flare; PCV-430/HV-430 N2 from the 700 kPa g header; HV-415 drain to closed drain	Instruments	LT-400/LIC-400, LG-400, PT-420/PIC-420, TI-410, AI-400, LSHH-416, LSL-417, PSHH-426 (own sensors, SIS I-1/I-2)
Drawing	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Document	FD-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Level	MORE	High level	- LCV-400 fails closed - HV-411 gate drops - XV-418 closes spuriously - LT-400 freezes or reads low - receiver pressure high - feed above design - hot oil lost (less flash, more liquid)	Liquid reaches the mesh pad and vapour nozzle: liquid slugs into the vapour line and fuel-gas header (burner flooding and fire at users); vapour space lost, pressure rises.	- LIC-400 HI alarm 80 % - LG-400 - LSHH-416 (own sensor) > 90 % for 5 s → I-1 closes XV-408 and XV-409	4	3	12 H	R-19, R-20	Instrument / Operations · Process / Instrument
N4.2	Level	LESS	Low level (liquid seal lost)	- LCV-400 fails open - HV-413 bypass left open - LIC-400 in MAN at a high output - LT-400 reads high - HV-415 drain left open	Vapour blows through LCV-400 into the downstream receiver built for liquid at 40 kPa g: receiver over-pressure, rupture, release; downstream pumps lose suction.	- LIC-400 LO alarm 20 % - LSLL-417 (own sensor) < 5 % for 5 s → I-2 closes XV-418 - LCV-400 fail closed	5	2	10 H	R-20, R-21	Process · Process / Instrument
N4.3	Pressure	MORE	High drum pressure	- Vapour outlet blocked (HV-421 gate drops, PCV-420 stuck) - header back-pressure - PT-420 reads low - feed over-heated - N2 regulator PCV-430 fails open from the 700 kPa g header	Drum pressure above MAWP (450 kPa g): rupture and large release of C4/C6, fire or explosion; liquid pushed out faster to the receiver.	- PIC-420 HI alarm 130 kPa g - PSHH-426 (own sensor) > 250 kPa g for 3 s → I-1 - PSV-400 set 450 kPa g to flare	5	2	10 H	R-14, R-15, R-17	Instrument / Operations · Process
N4.4	Pressure	LESS	Low pressure or vacuum	- Hot oil lost (no flash) and the liquid cools - PCV-420 fails open - PIC-420 set point low - PT-420 reads high - N2 supply lost (HV-430 closed)	More flash and a colder drum (auto-refrigeration); without N2 the drum can fall below atmospheric: collapse if not vacuum-rated, air in-leakage at flanges → flammable mixture inside.	- PIC-420 LO alarm 70 kPa g - PCV-430 N2 blanket 30 kPa g - TI-410 LO alarm 40 °C	4	2	8 M	R-16	Process / Mechanical
N4.5	Temperature	MORE	External fire at D-400	Pool fire from a leak at a flange, drain or valve station below the drum	Liquid boils off, the unwetted shell above the liquid overheats and fails below the PSV set pressure; large fire or vessel rupture.	- PSV-400 set 450 kPa g (fire case to be verified) - PSHH-426 I-1 stops feed and hot oil - closed drain	5	2	10 H	R-14, R-29	Process · Safety
N4.6	Composition	AS WELL AS	Liquid carried over with the vapour	- Foaming (surfactant in the feed after a tank switch) - level high - vapour rate above design (FCV-400 open, over-heated feed)	Liquid C6 in the vapour product: AI-420 below spec; liquid in the fuel-gas header, flame-out or flooding at the users' burners.	- AI-420 LO alarm 75 wt% - LIC-400 HI alarm - FI-420 HI alarm - LSHH-416	3	3	9 M	R-22, R-23	Mechanical / Process · Operations / Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.7	Separation	PART OF	Mesh pad removes only part of the droplets	Mesh pad damaged, displaced or plugged after an upset	Fine C6 mist in the fuel gas; condensate in the header low points; AI-420 slightly low.	- AI-420 LO alarm - header knock-out (off-plot)	2	3	6 M	R-22	Mechanical / Process
N4.8	Composition	AS WELL AS	Nitrogen in the vapour product	- PCV-430 regulator passing while the drum is above 30 kPa g - N2 sweep left on after start-up	Lower heating value of the fuel gas; burner instability at the header users; N2 wasted.	- AI-400 HI alarm 5 mol% - HV-430 isolation	2	3	6 M	R-15	Process
N4.9	Composition	OTHER THAN	Oxygen inside the drum	- Incomplete N2 sweep at start-up - air drawn in under vacuum - equipment opened and not purged	Flammable mixture inside D-400: internal explosion on ignition (static, pyrophoric scale).	- Start-up N2 sweep procedure - AI-400 (measures N2, not O2) - N2 blanket	5	2	10 H	R-26, R-16	Operations · Process / Mechanical
N4.10	Measurement	OTHER THAN	LT-400 reads wrong	- Transmitter freezes - impulse line plugged - turbulence or foam at the taps	LIC-400 moves LCV-400 on a false level: the true level can reach the vapour nozzle or empty unseen.	- LG-400 gauge glass (rounds) - LSHH-416 and LSL-417 have their own sensors	4	3	12 H	R-19	Instrument / Operations
N4.11	Measurement	OTHER THAN	PT-420 reads wrong	Transmitter drift (up to 40 kPa), impulse line fault	PIC-420 holds the wrong true pressure: products drift off spec; a transmitter reading low hides a real over-pressure.	- PSHH-426 own sensor - TI-410 against the bubble point	3	3	9 M	R-19	Instrument / Operations
N4.12	Relief	OTHER THAN	PSV-400 passing	- Seat leakage after a lift - set-point drift	Continuous product loss to flare (cost, flaring); the valve may not reseal after the next lift.	- PSV tail-pipe temperature (field) - material balance	2	3	6 M	R-31, R-14	Instrument · Process

Node N5 · Vapour line to the header

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Node	N5 · Vapour line to the header: D-400 top → FE-420 → PCV-420 → HV-421 → fuel-gas header
Design intent	Send 611 Nm ³ /h of vapour (76.8 wt% C4, spec ≥ 75) through PCV-420 (fail open, OUT 38.7 %) and HV-421 to the fuel-gas header at 30 kPa g, holding D-400 at 102.7 kPa g.	Design conditions	611 Nm ³ /h (1714 kg/h); 102.7 kPa g in the drum → 30 kPa g in the header
Equipment	Vapour line, FE-420, PCV-420 with block valves and globe bypass HV-423, HV-421 with bypass HV-422 (NC)	Instruments	PT-420/PIC-420, FI-420, AI-420; PSHH-426 on the drum top
Drawing	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Document	FD-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Flow	NO	No vapour to the header	- HV-421 gate drops (fails closed) - PCV-420 sticks closed - header isolated by others - HV-422 and HV-423 closed	Drum pressure rises at the flash rate (N4 pressure MORE): I-1 trip or PSV-400 lift to flare; liquid richer in C4 (AI-410 off spec) meanwhile.	- PIC-420 HI alarm 130 kPa g - FI-420 LO alarm 300 Nm³/h - PSHH-426 I-1 at 250 kPa g - PSV-400	4	3	12 H	R-17, R-30	Instrument / Operations · Operations
N5.2	Flow	MORE	More vapour to the header	- PCV-420 fails open (instrument air loss, positioner) - PIC-420 set point too low - HV-423 opened - feed over-heated or lighter	Drum depressures toward the header at several times the design vapour rate (see dynamic check): header pressure rises and upsets other users; liquid auto-refrigerates; heavier vapour.	- FI-420 HI alarm 900 Nm³/h - PIC-420 LO alarm 70 kPa g - TI-410 LO alarm - header pressure control (off-plot)	3	3	9 M	R-18	Process
N5.3	Pressure	MORE	Header back-pressure high	- Header users stop taking fuel gas - header pressure control fault	PCV-420 opens wide; once it saturates the drum pressure rises with the header: less flash, AI-410 above spec; back-flow if the header passes the drum pressure.	- PIC-420 HI alarm - AI-410 HI alarm - PSHH-426 - header relief (off-plot)	3	3	9 M	R-17	Instrument / Operations
N5.4	Flow	REVERSE	Header gas flows back into D-400	Drum pressure below the header (30 kPa g): hot oil lost with the blanket at 30 kPa g, drum depressured for maintenance with HV-421 open	Header gas (possibly with air if the header is being worked on) enters D-400; the drum cannot be isolated or purged.	- None: no check valve in the vapour line on the P&ID - HV-421 manual	3	3	9 M	R-05	Process
N5.5	Composition	LESS	Vapour below spec (AI-420 < 75 wt%)	- Feed heavier - feed over-heated - drum pressure low - carry-over	Off-spec fuel gas sold at a lower price; heavier gas condenses in the header (liquid in the fuel-gas system).	AI-420 LO alarm 75 wt%	2	4	8 M	R-22, R-23	Mechanical / Process · Operations / Instrument
N5.6	Temperature	LESS	Vapour line cooled (condensation)	- Rain or cold ambient on an uninsulated line - heavy vapour	Condensate collects in low points and at PCV-420; liquid slugs reach the header.	- Line routing (to confirm) - none on the P&ID	2	3	6 M	R-22	Mechanical / Process
N5.7	Integrity	OTHER THAN	Leak at the PCV-420 station	- Gasket or packing failure - vibration from choked flow across PCV-420	Flammable gas released at about 102.7 kPa g: flash fire or jet fire in the unit.	- None on the P&ID (no gas detection) - field rounds	4	2	8 M	R-29	Safety

Node N6 · Liquid line to the downstream unit

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Node	N6 · Liquid line to the downstream unit: D-400 bottom → FE-410 → XV-418 → LCV-400 → HV-411 → receiver
Design intent	Send 14.0 m ³ /h of liquid (26.6 wt% C4, spec ≤ 28) through XV-418 (fail closed), LCV-400 (fail closed, OUT 63.4 %) and HV-411 to the downstream receiver at 40 kPa g, driven by drum pressure plus static head.	Design conditions	14.0 m ³ /h (8495 kg/h) at 49.2 °C; 102.7 kPa g plus static head → 40 kPa g
Equipment	Bottoms line, FE-410, XV-418 (SIS I-2), LCV-400 with block valves and globe bypass HV-413, HV-411 with bypass HV-412 (NC)	Instruments	FT-410/FIC-410 (Rev B), AI-410, TI-410; LSL-417 acts through I-2
Drawing	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Document	FD-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Flow	NO	No liquid outflow	- LCV-400 fails closed (positioner or air failure) - HV-411 gate drops - XV-418 closes spuriously - receiver isolated	D-400 level rises (N4 level MORE) to the high-high switch within minutes (see dynamic check); downstream unit starved.	- FIC-410 LO alarm 8 m³/h - LIC-400 HI alarm 80 % - LSHH-416 I-1	4	3	12 H	R-19, R-30	Instrument / Operations · Operations
N6.2	Flow	MORE	More liquid than made; gas blow-by	- LCV-400 fails open - HV-413 bypass open - LIC-400 in MAN at a high output - LT-400 reads high	The drum drains; once the level is lost, vapour at drum pressure blows through into the receiver built for liquid at 40 kPa g: over-pressure, release.	- FIC-410 HI alarm 18 m³/h - LIC-400 LO alarm - LSLL-417 I-2 closes XV-418 - receiver relief (off-plot)	5	2	10 H	R-20, R-21	Process · Process / Instrument
N6.3	Flow	REVERSE	Back-flow from the receiver into D-400	Receiver pressure above drum pressure plus head (drum at the N2 blanket after hot-oil loss or depressuring; receiver pressure swings)	Receiver contents flow into D-400; level rises unexpectedly; contamination.	- XV-418 closes only on low level - no check valve on the bottoms line	3	2	6 M	R-04	Process / Mechanical
N6.4	Pressure	MORE	Downstream pressure high or swinging	Receiver pressure control fault or cycling	Bottoms flow falls or swings at a fixed LCV-400 opening; level rises; downstream upset.	- LIC-400 - Rev B FIC-410 - FIC-410 alarms	2	4	8 M	R-10	Instrument / Operations
N6.5	Composition	MORE	Liquid product too light (AI-410 > 28 wt%)	- Feed lighter - hot feed too cold - drum pressure too high	Off-spec liquid to the downstream unit (lower price); light ends flash in the receiver and raise its pressure.	- AI-410 HI alarm 28 wt% - TI-410 LO alarm 40 °C	2	4	8 M	R-23	Operations / Instrument
N6.6	Temperature	LESS	Very cold liquid	Drum depressured quickly (PCV-420 fails open); auto-refrigeration	Liquid colder than normal in the bottoms line (the model gives about −0.5 °C at the lowest, far above a typical −29 °C minimum design metal temperature): ice and condensation on the line, valves stiff, off-spec liquid; brittle fracture is not credible on this unit.	TI-410 LO alarm 40 °C	3	2	6 M	R-25	Process / Operations
N6.7	Measurement	OTHER THAN	AI-410 frozen or wrong	- Analyzer sample line blocked - analyzer fault	Off-spec liquid not seen; the operator optimises on a false reading.	- Lab samples - TI-410 as an inferential check	2	3	6 M	R-23	Operations / Instrument

Node N7 · Procedures

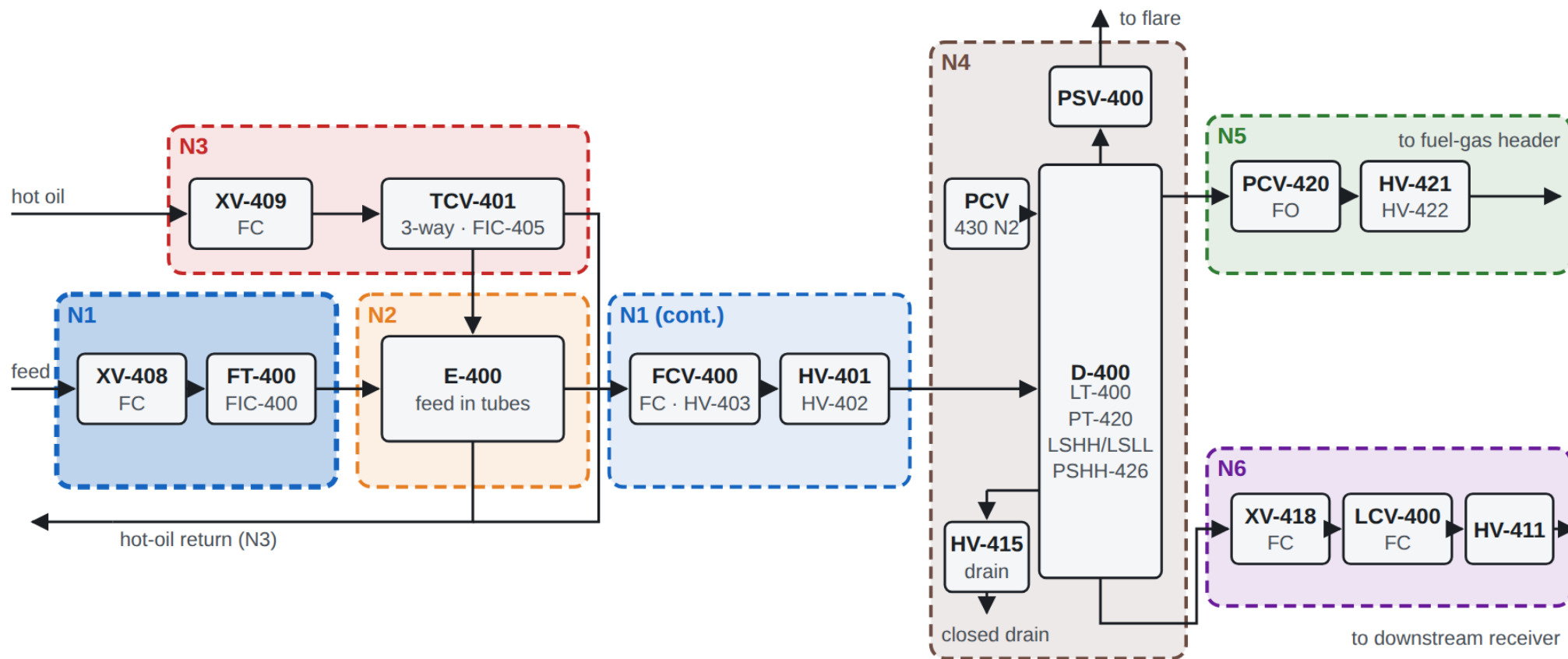
Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Node	N7 · Procedures: start-up with N2 sweep, shutdown, depressuring and draining, maintenance
Design intent	Start-up: sweep D-400 with N2, pressurise on the blanket, establish feed, then hot oil, then line out. Shutdown: hot oil off, feed off, depressure to the header, drain to the closed drain. Maintenance: isolate, depressure, purge, blind.	Design conditions	Procedure steps, targets and time limits in the Exercises tab
Equipment	Whole unit; SIS reset and permissives; HV-415 drain; PCV-430/HV-430	Instruments	AI-400; SIS reset permissives (level below 85 %, pressure below 230 kPa g; XV-418 only above 10 % level); procedures in the Exercises tab
Drawing	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Document	FD-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N7.1	Purge (N2 sweep)	LESS	N2 sweep incomplete	- Sweep time cut short - vented from one nozzle only (dead legs: LG-400, instrument nozzles, bottoms line) - AI-400 read before it settles	Air left in D-400 when hydrocarbon enters: flammable mixture, internal explosion on ignition.	- Procedure hold point - AI-400 (N2 only)	5	2	10 H	R-26	Operations
N7.2	Sequence	BEFORE	Hot oil before feed flow	- Step order not followed - TCV-401 or XV-409 left open from the last run	Stagnant feed in the E-400 tubes heated: vaporisation and blocked-in over-pressure (N2 pressure MORE).	Procedure: TIC-401 in MAN at 0 % and XV-409 closed until feed flows	4	2	8 M	R-01, R-27	Operations · Process / Instrument
N7.3	Sequence	AFTER	Hot oil stopped after the feed at shutdown	- Wrong order - XV-409 left open	As above: blocked-in feed heated in E-400.	- Procedure - TIC-401 moves TCV-401 to bypass on a high TT-401	4	2	8 M	R-01, R-27	Operations · Process / Instrument
N7.4	Timing	EARLY	Liquid sent downstream too early	- XV-418 opened or LIC-400 put in AUTO before the level is established - product sent before AI-410 is on spec	Gas blow-by to the receiver at low level; off-spec liquid downstream.	- XV-418 permissive: level above 10 % (SIS) - LSLL-417 I-2	3	2	6 M	R-27	Operations
N7.5	Timing	LATE	Late response to the I-2 closure	- XV-418 closed by LSLL-417 while the feed keeps running - alarm not acted on	Level climbs from low-low to the high-high switch: I-1 trips the unit (see dynamic check).	- LIC-400 HI alarm 80 % - LSHH-416 I-1	3	3	9 M	R-20, R-30	Operations · Process / Instrument
N7.6	Feed rate	LESS	Start-up held at a very low feed rate	Feed below the TIC-401 stable range for a long time	TIC-401 cycles; hot-feed peaks approach the bubble point; the drum pressure and vapour rate swing.	- Procedure feed step (Exercises tab) - TIC-401 HI alarm	2	3	6 M	R-10	Instrument / Operations
N7.7	Temperature (drain)	LESS	Auto-refrigeration while draining	- Liquid drained under pressure through HV-415 to the closed drain - C4 flashes across the valve	Valve and drain line chilled toward the boiling point of butane (about −0.5 °C at atmospheric pressure): ice on the valve can jam HV-415 open and release C4; cold burns. The model does not reach the carbon-steel minimum design metal temperature (typical −29 °C).	- HV-415 normally closed - closed drain - procedure	4	3	12 H	R-25	Process / Operations
N7.8	Depressuring	MORE	Depressuring too fast to the header	PCV-420 opened wide or HV-423 opened at shutdown	Auto-refrigeration of the drum liquid; header overload; liquid carry-over.	- FI-420 HI alarm 900 Nm³/h - TI-410 LO alarm	3	3	9 M	R-25, R-18	Process · Process / Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N7.9	Maintenance	OTHER THAN	Isolation incomplete	- Gate valves passing (HV-401, HV-411, HV-421) - XV-408/409/418 passing - no blinds - drain or vent not proven - hot oil not isolated	Hydrocarbon or hot oil released when equipment is opened: fire, burns, exposure.	- Permit to work - lock-out	5	2	10 H	R-27	Operations
N7.10	Safety system	OTHER THAN	SIS bypassed or reset with the cause present	- Bypass left after testing - forced reset	High level or high pressure unprotected.	- Reset permissives (level below 85 %, pressure below 230 kPa g) - XV-418 permissive above 10 %	5	2	10 H	R-28	Instrument / Operations

Appendix 2 · HAZOP documents (drawings with node boundaries)

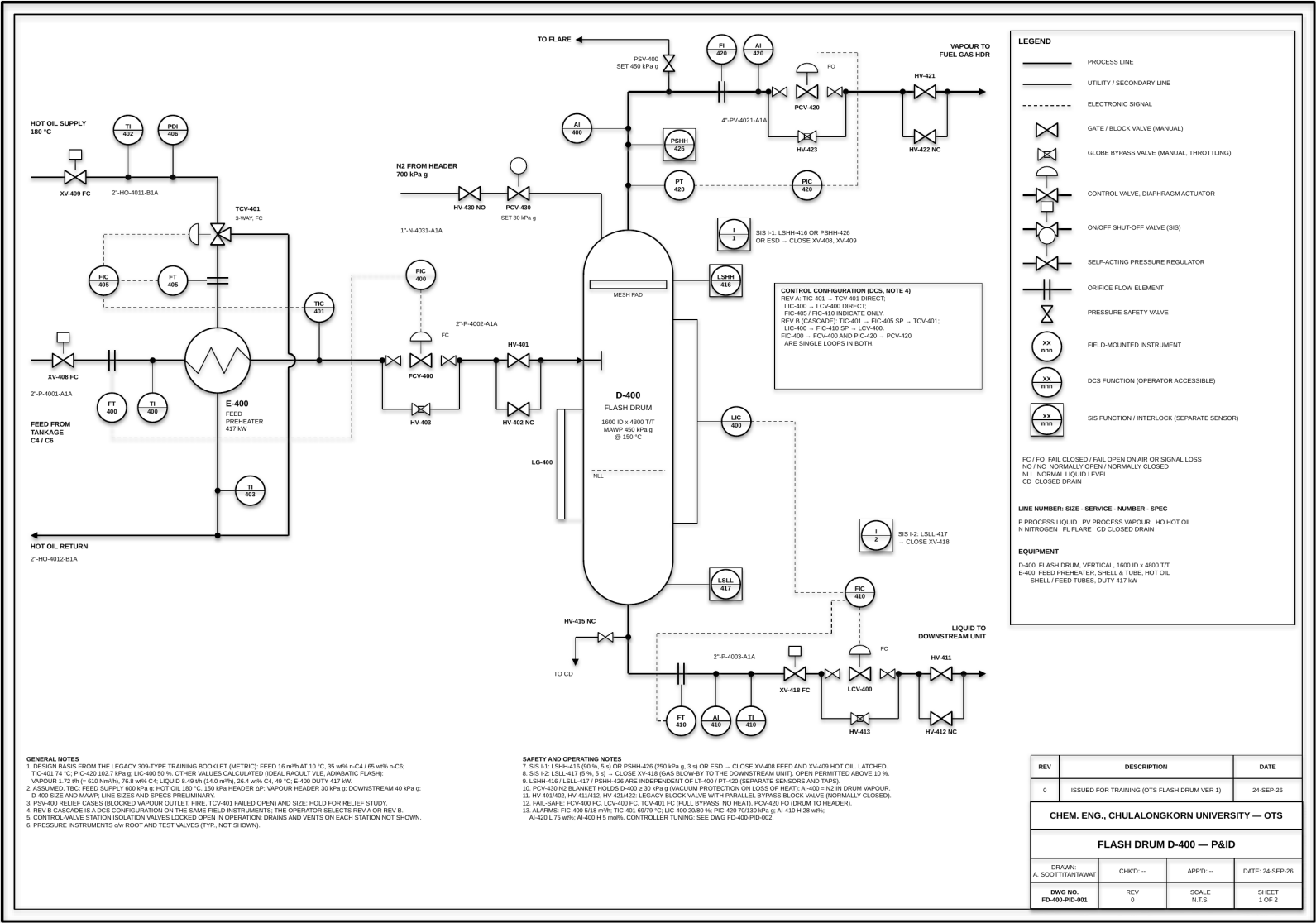
A2-1 · HAZOP node map



N7 (not drawn): start-up with N2 sweep, shutdown, depressuring and draining, maintenance

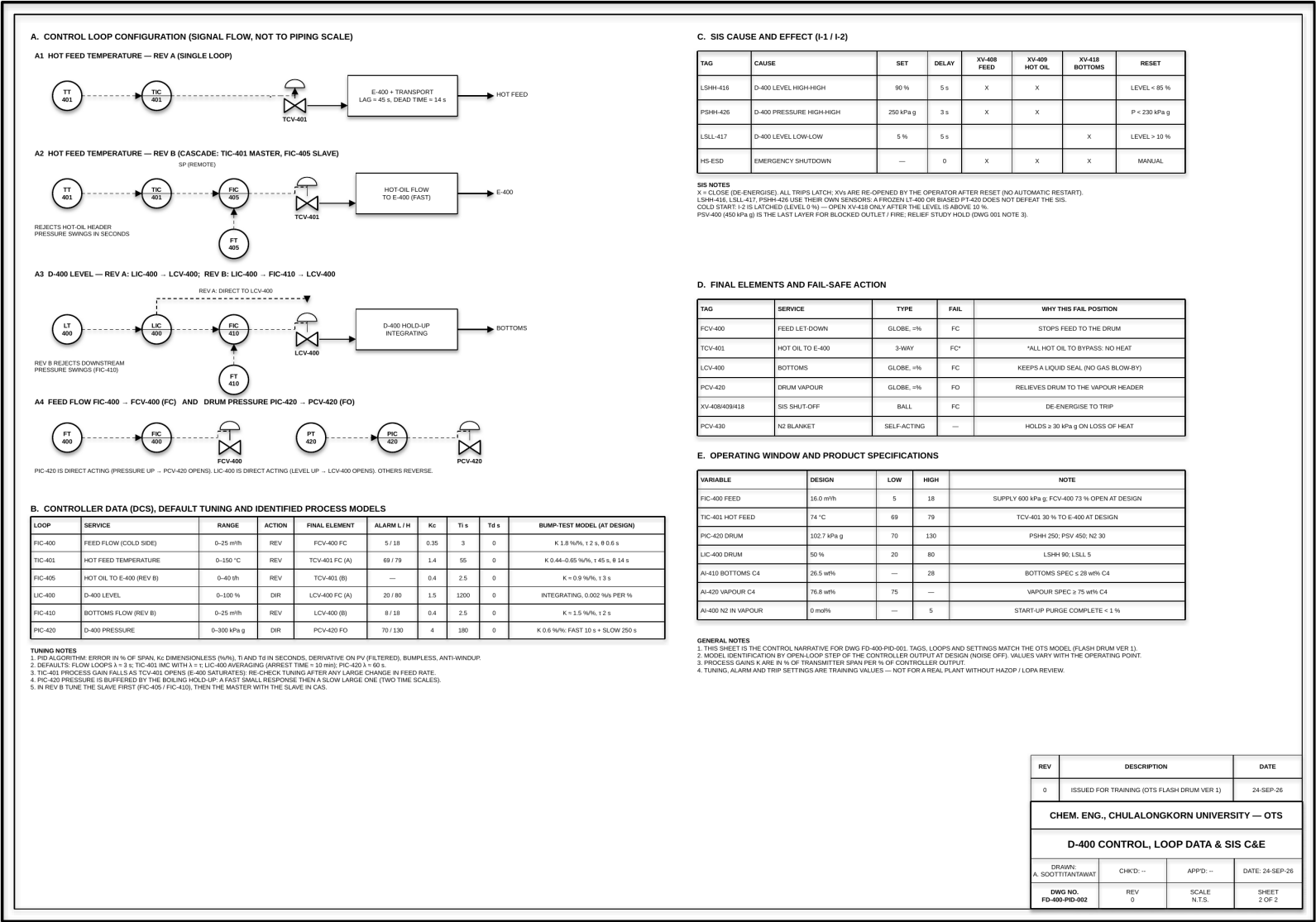
A2-1. HAZOP node map of D-400 (nodes N1-N6; N7 = procedures).

A2-2 · FD-400-PID-001 (piping and instruments)



P&ID sheet 1 of the D-400 unit, as shown on the P&ID tab of the simulator.

A2-3 · FD-400-PID-002 (control loops, loop data and SIS cause and effect)



P&ID sheet 2 of the D-400 unit: control scheme, controller data and the SIS cause and effect.

Appendix 3 · HAZOP action close-out sheets

One sheet per recommendation (31 sheets). The action party writes the response, ticks the status and signs; the HAZOP leader accepts the close-out.

HAZOP ACTION CLOSE-OUT SHEET · R-01

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-01
Node	N1/N2 · Feed supply and flow control / E-400 feed preheater, feed (tube) side	Action party	Process / Instrument
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow NO : No feed to D-400	- HV-401 feed block valve gate drops (fails closed) - FCV-400 shut (FIC-400 fault, positioner or instrument-air failure: fail closed) - XV-408 closes spuriously - supply pump trips upstream	Production stops. E-400 tubes hold stagnant feed while hot oil still flows: the feed heats toward the hot-oil temperature, vaporises and, if blocked in, over-pressures the tube side. D-400 level and pressure fall toward the N2 blanket.	- FIC-400 LO alarm 5 m³/h - TIC-401 HI alarm 79 °C and TIC-401 moves TCV-401 to bypass - LIC-400 closes LCV-400 - PCV-430 N2 blanket 30 kPa g	S3 L3 9 Medium
N1.5	Pressure MORE : Feed line or E-400 tubes over-pressured	- FCV-400, HV-403 and HV-401 closed with the supply pump running (shut-off 650 kPa g) - liquid blocked in between XV-408 and FCV-400 and heated by hot oil	Shut-off head alone is within the line rating, but blocked-in liquid heated by hot oil rises in pressure without limit: flange leak or tube rupture, release and fire.	- No relief on the E-400 tube side on the P&ID - TIC-401 cuts TCV-401 when TT-401 sees the heat (not when flow is zero at the element)	S4 L2 8 Medium
N2.5	Pressure MORE : Tube side blocked in while hot oil flows	- Feed stopped by FCV-400 or HV-401 (not by I-1) with XV-409 open or passing - shutdown in the wrong order	Trapped liquid expands and vaporises: tube-side pressure far above design, tube or flange failure, feed into the hot oil or to atmosphere, fire.	- SIS I-1 closes XV-409 with XV-408 (not on FCV-400 or HV-401 closure) - TIC-401 moves TCV-401 to bypass when TT-401 rises - no tube-side thermal relief on the P&ID	S4 L3 12 High
N7.2	Sequence BEFORE : Hot oil before feed flow	- Step order not followed - TCV-401 or XV-409 left open from the last run	Stagnant feed in the E-400 tubes heated: vaporisation and blocked-in over-pressure (N2 pressure MORE).	Procedure: TIC-401 in MAN at 0 % and XV-409 closed until feed flows	S4 L2 8 Medium
N7.3	Sequence AFTER : Hot oil stopped after the feed at shutdown	- Wrong order - XV-409 left open	As above: blocked-in feed heated in E-400.	- Procedure - TIC-401 moves TCV-401 to bypass on a high TT-401	S4 L2 8 Medium

HAZOP recommendation

Interlock the hot oil to the feed: close XV-409 on low-low feed flow (an element independent of FT-400) and whenever XV-408 closes, so E-400 is never heated with the tube side blocked in or stagnant.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-02

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-02
Node	N1/N2 · Feed supply and flow control / E-400 feed preheater, feed (tube) side	Action party	Process
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow NO : No feed to D-400	- HV-401 feed block valve gate drops (fails closed) - FCV-400 shut (FIC-400 fault, positioner or instrument-air failure: fail closed) - XV-408 closes spuriously - supply pump trips upstream	Production stops. E-400 tubes hold stagnant feed while hot oil still flows: the feed heats toward the hot-oil temperature, vaporises and, if blocked in, over-pressures the tube side. D-400 level and pressure fall toward the N2 blanket.	- FIC-400 LO alarm 5 m³/h - TIC-401 HI alarm 79 °C and TIC-401 moves TCV-401 to bypass - LIC-400 closes LCV-400 - PCV-430 N2 blanket 30 kPa g	S3 L3 9 Medium
N1.5	Pressure MORE : Feed line or E-400 tubes over-pressured	- FCV-400, HV-403 and HV-401 closed with the supply pump running (shut-off 650 kPa g) - liquid blocked in between XV-408 and FCV-400 and heated by hot oil	Shut-off head alone is within the line rating, but blocked-in liquid heated by hot oil rises in pressure without limit: flange leak or tube rupture, release and fire.	- No relief on the E-400 tube side on the P&ID - TIC-401 cuts TCV-401 when TT-401 sees the heat (not when flow is zero at the element)	S4 L2 8 Medium
N2.5	Pressure MORE : Tube side blocked in while hot oil flows	- Feed stopped by FCV-400 or HV-401 (not by I-1) with XV-409 open or passing - shutdown in the wrong order	Trapped liquid expands and vaporises: tube-side pressure far above design, tube or flange failure, feed into the hot oil or to atmosphere, fire.	- SIS I-1 closes XV-409 with XV-408 (not on FCV-400 or HV-401 closure) - TIC-401 moves TCV-401 to bypass when TT-401 rises - no tube-side thermal relief on the P&ID	S4 L3 12 High

HAZOP recommendation

Fit a thermal relief valve on the E-400 tube side (between XV-408 and FCV-400), sized for blocked-in liquid heated by hot oil at 180 °C, discharging to a closed system.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-03

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-03
Node	N1 · Feed supply and flow control	Action party	Process / Mechanical
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.4	Flow REVERSE : Back-flow from E-400 and D-400 toward the supply	- Supply pump stops with D-400 at pressure and XV-408 open - supply pressure below drum pressure during start-up	Hot, light hydrocarbon flows back into the supply line and tank; flashing and hammer in the supply line; D-400 inventory lost.	- FIC-400 LO alarm (FIC-400 then opens FCV-400 further, which helps the back-flow) - XV-408 closes only on SIS I-1	S3 L2 6 Medium

HAZOP recommendation

Add a check valve in the feed line downstream of XV-408 so D-400 and E-400 cannot flow back into the supply when the supply pump stops.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-04

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-04
Node	N6 · Liquid line to the downstream unit	Action party	Process / Mechanical
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.3	Flow REVERSE : Back-flow from the receiver into D-400	Receiver pressure above drum pressure plus head (drum at the N2 blanket after hot-oil loss or depressuring; receiver pressure swings)	Receiver contents flow into D-400; level rises unexpectedly; contamination.	• XV-418 closes only on low level • no check valve on the bottoms line	S3 L2 6 Medium

HAZOP recommendation

Add a check valve in the bottoms line downstream of HV-411 against back-flow from the downstream receiver (40 kPa g) when D-400 pressure is low.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-05

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-05
Node	N5 · Vapour line to the header	Action party	Process
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.4	Flow REVERSE : Header gas flows back into D-400	Drum pressure below the header (30 kPa g): hot oil lost with the blanket at 30 kPa g, drum depressured for maintenance with HV-421 open	Header gas (possibly with air if the header is being worked on) enters D-400; the drum cannot be isolated or purged.	• None: no check valve in the vapour line on the P&ID • HV-421 manual	S3 L3 9 Medium

HAZOP recommendation

Add a check valve in the vapour line (or a low-pressure closure of HV-421 by procedure) so header gas cannot flow back into D-400 when the drum falls to the N2 blanket pressure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-06

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-06
Node	N1 · Feed supply and flow control	Action party	Instrument
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.2	Flow LESS : Less feed than set	- Supply pressure falls (upstream pump loses head) until FCV-400 runs out of travel - FT-400 reads high (drift, plugged impulse line) - FCV-400 stiction after packing work - HV-401 partly closed	Lower throughput. At low flow the E-400 outlet runs hotter for the same oil, TIC-401 gain and dead time grow and the loop cycles; flow swings reach the drum as vapour-rate swings.	- FIC-400 LO alarm 5 m³/h - FIC-400 opens FCV-400 - TIC-401 HI/LO alarms	S2 L4 8 Medium
N1.6	Pressure LESS : Low supply pressure	- Upstream pump losing head - low supply-tank level - plugged strainer	FCV-400 opens to hold the flow and loses its margin unseen; lower pressure in E-400 lowers the bubble point of the hot feed (vapour ahead of FCV-400).	- FIC-400 LO alarm (only after FCV-400 saturates) - TIC-401	S2 L3 6 Medium

HAZOP recommendation

Add a feed supply pressure indicator with low alarm upstream of FCV-400 and a FIC-400 output-high alarm, so loss of supply head is seen before FCV-400 runs out of travel.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-07

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-07
Node	N1 · Feed supply and flow control	Action party	Process
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.3	Flow MORE : More feed than design	- FCV-400 fails wide open (positioner fault, FIC-400 in MAN at a high output) - FT-400 reads low - set point entered too high - globe bypass HV-403 opened with FCV-400 in service	E-400 cannot heat the extra feed (hot feed colder, less vapour per tonne) while the total vapour and liquid rates rise well above design (see dynamic check): level rises, mesh-pad overload and carry-over, header and downstream receiver overloaded.	- FIC-400 HI alarm 20 m³/h - LIC-400 opens LCV-400 - LIC-400 HI alarm 80 % - FI-420 HI alarm 900 Nm³/h - LSHH-416 I-1 at 90 %	S3 L3 9 Medium

HAZOP recommendation

Limit the feed with FCV-400 wide open (travel stop or restriction orifice), or show that D-400, the mesh pad, PCV-420 and the downstream receiver handle that flow.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-08

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-08
Node	N2/N3 · E-400 feed preheater, feed (tube) side / Hot-oil side of E-400	Action party	Process / Instrument
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.1	Temperature MORE : Hot feed above set point	• TCV-401 driven to full E-400 (TIC-401 in MAN at a high output, positioner fault) • set point entered too high • FIC-405 fault in Rev B	Hot feed approaches or passes its bubble point at the FCV-400 inlet: vapour forms in E-400 and ahead of FCV-400 (two-phase let-down, vibration, flow and level swings); more and heavier vapour (AI-420 off spec); PCV-420 and header overloaded; drum pressure rises.	• TIC-401 HI alarm 79 °C • TI-410 HI alarm 60 °C • AI-420 LO alarm • PIC-420 opens PCV-420 • PSHH-426 I-1 at 250 kPa g • PSV-400	S3 L3 9 Medium
N2.3	Measurement OTHER THAN : TT-401 reads low	• Transmitter drift or fault • deposits on the thermowell • wrong calibration	TIC-401 over-heats the feed while the display shows the set point: the true hot-feed temperature rises toward its bubble point at the FCV-400 inlet (see dynamic check); more, heavier vapour; lighter liquid.	• TI-410 HI alarm 60 °C • AI-420 LO alarm 75 wt% • TI-403 rises (indirect) • no second element on the E-400 outlet	S3 L3 9 Medium
N3.3	Flow MORE : More hot oil than needed	• TCV-401 fails to full E-400 (positioner) or TIC-401 in MAN at a high output • header ΔP high	Hot feed over-heated: vapour ahead of FCV-400 (see N2 temperature MORE); PSV-400 relief case "TCV-401 failed open".	• PDI-406 HI alarm 200 kPa • TIC-401 HI alarm 79 °C • PSHH-426 • PSV-400	S3 L3 9 Medium

HAZOP recommendation

Evaluate by LOPA an independent high-temperature trip on the E-400 outlet (own element) closing XV-409 below the bubble point of the feed at the FCV-400 inlet pressure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-09

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-09
Node	N2 · E-400 feed preheater, feed (tube) side	Action party	Instrument
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.3	Measurement OTHER THAN : TT-401 reads low	• Transmitter drift or fault • deposits on the thermowell • wrong calibration	TIC-401 over-heats the feed while the display shows the set point: the true hot-feed temperature rises toward its bubble point at the FCV-400 inlet (see dynamic check); more, heavier vapour; lighter liquid.	• TI-410 HI alarm 60 °C • AI-420 LO alarm 75 wt% • TI-403 rises (indirect) • no second element on the E-400 outlet	S3 L3 9 Medium

HAZOP recommendation

Add a second temperature element at the E-400 outlet with a deviation alarm against TT-401.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-10

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-10
Node	N2/N3/N6 · E-400 feed preheater, feed (tube) side / Hot-oil side of E-400 / Liquid line to the downstream unit	Action party	Instrument / Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.2	Flow LESS : Less feed than set	- Supply pressure falls (upstream pump loses head) until FCV-400 runs out of travel - FT-400 reads high (drift, plugged impulse line) - FCV-400 stiction after packing work - HV-401 partly closed	Lower throughput. At low flow the E-400 outlet runs hotter for the same oil, TIC-401 gain and dead time grow and the loop cycles; flow swings reach the drum as vapour-rate swings.	- FIC-400 LO alarm 5 m³/h - FIC-400 opens FCV-400 - TIC-401 HI/LO alarms	S2 L4 8 Medium
N2.7	Control OTHER THAN : Hot-feed temperature cycling	- Feed rate near the TIC-401 cycling limit - hot-oil header pressure swings in Rev A - poor tuning	Hot-feed swings move the vapour rate and drum pressure; at the peaks, vapour forms ahead of FCV-400; product quality swings.	- TIC-401 HI/LO alarms - Rev B cascade TIC-401 → FIC-405 - tuning records (Tuning lab)	S2 L4 8 Medium
N3.2	Flow LESS : Less hot oil than demanded	- Header ΔP falls or swings (other users) - TCV-401 sticking - FT-405 reads high (Rev B)	TCV-401 opens further; once it saturates TIC-401 falls (temperature LESS); header swings cycle TIC-401 in Rev A.	- PDI-406 LO alarm 100 kPa - TIC-401 LO alarm - Rev B FIC-405 rejects header swings	S2 L4 8 Medium
N6.4	Pressure MORE : Downstream pressure high or swinging	Receiver pressure control fault or cycling	Bottoms flow falls or swings at a fixed LCV-400 opening; level rises; downstream upset.	- LIC-400 - Rev B FIC-410 - FIC-410 alarms	S2 L4 8 Medium
N7.6	Feed rate LESS : Start-up held at a very low feed rate	Feed below the TIC-401 stable range for a long time	TIC-401 cycles; hot-feed peaks approach the bubble point; the drum pressure and vapour rate swing.	- Procedure feed step (Exercises tab) - TIC-401 HI alarm	S2 L3 6 Medium

HAZOP recommendation

Run the Rev B cascades (TIC-401 → FIC-405, LIC-400 → FIC-410) as the normal configuration; state the minimum stable feed rate for TIC-401 in the start-up procedure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-11

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-11
Node	N3 · Hot-oil side of E-400	Action party	Instrument / Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.2	Temperature LESS : Hot feed below set point	- Hot-oil failure - hot-oil supply cooler - E-400 fouling - TCV-401 fails to full bypass (air or signal loss) - feed above E-400 capacity	Little or no flash: vapour product falls, drum pressure falls to the N2 blanket and can fall below it, level rises; bottoms far above the 28 wt% spec sent downstream.	- TIC-401 LO alarm 69 °C - TI-410 LO alarm 40 °C - PIC-420 LO alarm 70 kPa g - AI-410 HI alarm - PCV-430 N2 blanket 30 kPa g	S3 L4 12 High
N3.1	Flow NO : No hot oil to E-400	- Hot-oil circulation pump trips off-site - XV-409 closes spuriously - TCV-401 fails to full bypass (instrument air or signal loss)	Feed reaches D-400 near its supply temperature: no flash, vapour product stops, drum pressure falls to the N2 blanket and below, level rises, bottoms far off spec.	- PDI-406 LO alarm 100 kPa (header failure only) - TIC-401 LO alarm - TI-410 LO alarm - PIC-420 LO alarm - PCV-430 N2 blanket	S3 L3 9 Medium
N3.2	Flow LESS : Less hot oil than demanded	- Header ΔP falls or swings (other users) - TCV-401 sticking - FT-405 reads high (Rev B)	TCV-401 opens further; once it saturates TIC-401 falls (temperature LESS); header swings cycle TIC-401 in Rev A.	- PDI-406 LO alarm 100 kPa - TIC-401 LO alarm - Rev B FIC-405 rejects header swings	S2 L4 8 Medium
N3.4	Temperature MORE : Hot oil hotter than 180 °C	Hot-oil heater control fault off-site	At low feed the outlet can approach the oil temperature, far above the bubble point; film boiling and faster fouling on the tube wall.	- TI-402 indication only (no alarm) - TIC-401 closes TCV-401	S2 L3 6 Medium
N3.5	Temperature LESS : Hot-oil supply cooler	Hot-oil heater cuts back or trips	TCV-401 opens and FI-405 rises to hold TIC-401, leaving little valve margin; a further upset leaves TIC-401 below set point.	- TIC-401 LO alarm 69 °C - TI-402 indication (no alarm)	S2 L4 8 Medium

HAZOP recommendation

Add hot-oil supply alarms: TI-402 low and high, FI-405 low; agree with utilities to warn the unit before hot-oil changes.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-12

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-12
Node	N2 · E-400 feed preheater, feed (tube) side	Action party	Process / Instrument
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.2	Temperature LESS : Hot feed below set point	- Hot-oil failure - hot-oil supply cooler - E-400 fouling - TCV-401 fails to full bypass (air or signal loss) - feed above E-400 capacity	Little or no flash: vapour product falls, drum pressure falls to the N2 blanket and can fall below it, level rises; bottoms far above the 28 wt% spec sent downstream.	- TIC-401 LO alarm 69 °C - TI-410 LO alarm 40 °C - PIC-420 LO alarm 70 kPa g - AI-410 HI alarm - PCV-430 N2 blanket 30 kPa g	S3 L4 12 High
N2.4	Heat transfer LESS : E-400 fouling	Deposits from the feed (tank sludge) or hot-oil breakdown products on the shell side	TCV-401 opens slowly to 100 %, then TIC-401 falls below set point (as temperature LESS); more hot oil used.	- TIC-401 LO alarm 69 °C - TI-403 trend	S2 L4 8 Medium

HAZOP recommendation

Add an E-400 performance display (UA from TI-400, TT-401, TI-402, TI-403 and FI-405) and a TCV-401 output-high alarm; plan cleaning on the trend.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-13

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-13
Node	N2/N3 · E-400 feed preheater, feed (tube) side / Hot-oil side of E-400	Action party	Mechanical / Process
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.6	Integrity OTHER THAN: E-400 tube leak	• Tube corrosion, erosion or vibration • thermal shock (cold feed onto hot tubes at start-up)	Feed into the hot oil (feed side ≈ 550 kPa g in normal operation): C4 flashes in the hot-oil return and expansion vessel of other users, fire. Hot oil into the feed (when the feed side is depressured): heavy oil in the D-400 bottoms, off-spec liquid, fouling downstream.	• None specific • AI-410 • hot-oil system relief (off-plot, not on the P&ID)	S4 L2 8 Medium
N3.6	Pressure MORE: Hot-oil side over-pressured	• E-400 tube rupture lets feed (≈ 550 kPa g) into the shell • return blocked with the supply open	Shell and hot-oil return over-pressured; C4 carried into the hot-oil system (fire at the hot-oil heater or expansion vessel).	• Hot-oil system relief (off-plot, not on the P&ID) • PDI-406 HI alarm	S4 L2 8 Medium

HAZOP recommendation

Confirm the E-400 tube-rupture case: shell (hot-oil) design pressure against the tube side (two-thirds rule) and relief of the hot-oil system; test the hot oil for light ends (flash point) periodically.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-14

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-14
Node	N4 · D-400 flash drum	Action party	Process
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.1	Temperature MORE : Hot feed above set point	• TCV-401 driven to full E-400 (TIC-401 in MAN at a high output, positioner fault) • set point entered too high • FIC-405 fault in Rev B	Hot feed approaches or passes its bubble point at the FCV-400 inlet: vapour forms in E-400 and ahead of FCV-400 (two-phase let-down, vibration, flow and level swings); more and heavier vapour (AI-420 off spec); PCV-420 and header overloaded; drum pressure rises.	• TIC-401 HI alarm 79 °C • TI-410 HI alarm 60 °C • AI-420 LO alarm • PIC-420 opens PCV-420 • PSHH-426 I-1 at 250 kPa g • PSV-400	S3 L3 9 Medium
N3.3	Flow MORE : More hot oil than needed	• TCV-401 fails to full E-400 (positioner) or TIC-401 in MAN at a high output • header ΔP high	Hot feed over-heated: vapour ahead of FCV-400 (see N2 temperature MORE); PSV-400 relief case "TCV-401 failed open".	• PDI-406 HI alarm 200 kPa • TIC-401 HI alarm 79 °C • PSHH-426 • PSV-400	S3 L3 9 Medium
N4.3	Pressure MORE : High drum pressure	• Vapour outlet blocked (HV-421 gate drops, PCV-420 stuck) • header back-pressure • PT-420 reads low • feed over-heated • N2 regulator PCV-430 fails open from the 700 kPa g header	Drum pressure above MAWP (450 kPa g): rupture and large release of C4/C6, fire or explosion; liquid pushed out faster to the receiver.	• PIC-420 HI alarm 130 kPa g • PSHH-426 (own sensor) > 250 kPa g for 3 s → I-1 • PSV-400 set 450 kPa g to flare	S5 L2 10 High
N4.5	Temperature MORE : External fire at D-400	Pool fire from a leak at a flange, drain or valve station below the drum	Liquid boils off, the unwetted shell above the liquid overheats and fails below the PSV set pressure; large fire or vessel rupture.	• PSV-400 set 450 kPa g (fire case to be verified) • PSHH-426 I-1 stops feed and hot oil • closed drain	S5 L2 10 High
N4.12	Relief OTHER THAN : PSV-400 passing	• Seat leakage after a lift • set-point drift	Continuous product loss to flare (cost, flaring); the valve may not reseal after the next lift.	• PSV tail-pipe temperature (field) • material balance	S2 L3 6 Medium

HAZOP recommendation

Verify PSV-400 (450 kPa g) for every relief case: blocked vapour outlet, external fire, TCV-401 failed to full E-400 and PCV-430 failed open; record them in the relief study.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-15

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-15
Node	N4 · D-400 flash drum	Action party	Process
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.3	Pressure MORE : High drum pressure	- Vapour outlet blocked (HV-421 gate drops, PCV-420 stuck) - header back-pressure - PT-420 reads low - feed over-heated - N2 regulator PCV-430 fails open from the 700 kPa g header	Drum pressure above MAWP (450 kPa g): rupture and large release of C4/C6, fire or explosion; liquid pushed out faster to the receiver.	- PIC-420 HI alarm 130 kPa g - PSHH-426 (own sensor) > 250 kPa g for 3 s → I-1 - PSV-400 set 450 kPa g to flare	S5 L2 10 High
N4.8	Composition AS WELL AS : Nitrogen in the vapour product	- PCV-430 regulator passing while the drum is above 30 kPa g - N2 sweep left on after start-up	Lower heating value of the fuel gas: burner instability at the header users; N2 wasted.	- AI-400 HI alarm 5 mol% - HV-430 isolation	S2 L3 6 Medium

HAZOP recommendation

Fit a restriction orifice after PCV-430 so a failed-open regulator on the 700 kPa g N2 header cannot exceed PSV-400 capacity.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-16

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-16
Node	N4 · D-400 flash drum	Action party	Process / Mechanical
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.2	Temperature LESS : Hot feed below set point	- Hot-oil failure - hot-oil supply cooler - E-400 fouling - TCV-401 fails to full bypass (air or signal loss) - feed above E-400 capacity	Little or no flash: vapour product falls, drum pressure falls to the N2 blanket and can fall below it, level rises; bottoms far above the 28 wt% spec sent downstream.	- TIC-401 LO alarm 69 °C - TI-410 LO alarm 40 °C - PIC-420 LO alarm 70 kPa g - AI-410 HI alarm - PCV-430 N2 blanket 30 kPa g	S3 L4 12 High
N3.1	Flow NO : No hot oil to E-400	- Hot-oil circulation pump trips off-site - XV-409 closes spuriously - TCV-401 fails to full bypass (instrument air or signal loss)	Feed reaches D-400 near its supply temperature: no flash, vapour product stops, drum pressure falls to the N2 blanket and below, level rises, bottoms far off spec.	- PDI-406 LO alarm 100 kPa (header failure only) - TIC-401 LO alarm - TI-410 LO alarm - PIC-420 LO alarm - PCV-430 N2 blanket	S3 L3 9 Medium
N4.4	Pressure LESS : Low pressure or vacuum	- Hot oil lost (no flash) and the liquid cools - PCV-420 fails open - PIC-420 set point low - PT-420 reads high - N2 supply lost (HV-430 closed)	More flash and a colder drum (auto-refrigeration); without N2 the drum can fall below atmospheric: collapse if not vacuum-rated, air in-leakage at flanges → flammable mixture inside.	- PIC-420 LO alarm 70 kPa g - PCV-430 N2 blanket 30 kPa g - TI-410 LO alarm 40 °C	S4 L2 8 Medium
N4.9	Composition OTHER THAN : Oxygen inside the drum	- Incomplete N2 sweep at start-up - air drawn in under vacuum - equipment opened and not purged	Flammable mixture inside D-400: internal explosion on ignition (static, pyrophoric scale).	- Start-up N2 sweep procedure - AI-400 (measures N2, not O2) - N2 blanket	S5 L2 10 High

HAZOP recommendation

Confirm the D-400 vacuum rating, or the N2 blanket capacity, for the cold-drum cases (hot oil lost, cold feed, fast depressuring); add a PT-420 low-low alarm.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-17

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-17
Node	N4/N5 · D-400 flash drum / Vapour line to the header	Action party	Instrument / Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.3	Pressure MORE : High drum pressure	- Vapour outlet blocked (HV-421 gate drops, PCV-420 stuck) - header back-pressure - PT-420 reads low - feed over-heated - N2 regulator PCV-430 fails open from the 700 kPa g header	Drum pressure above MAWP (450 kPa g): rupture and large release of C4/C6, fire or explosion; liquid pushed out faster to the receiver.	- PIC-420 HI alarm 130 kPa g - PSHH-426 (own sensor) > 250 kPa g for 3 s → I-1 - PSV-400 set 450 kPa g to flare	S5 L2 10 High
N5.1	Flow NO : No vapour to the header	- HV-421 gate drops (fails closed) - PCV-420 sticks closed - header isolated by others - HV-422 and HV-423 closed	Drum pressure rises at the flash rate (N4 pressure MORE): I-1 trip or PSV-400 lift to flare; liquid richer in C4 (AI-410 off spec) meanwhile.	- PIC-420 HI alarm 130 kPa g - FI-420 LO alarm 300 Nm³/h - PSHH-426 I-1 at 250 kPa g - PSV-400	S4 L3 12 High
N5.3	Pressure MORE : Header back-pressure high	- Header users stop taking fuel gas - header pressure control fault	PCV-420 opens wide; once it saturates the drum pressure rises with the header: less flash, AI-410 above spec; back-flow if the header passes the drum pressure.	- PIC-420 HI alarm - AI-410 HI alarm - PSHH-426 - header relief (off-plot)	S3 L3 9 Medium

HAZOP recommendation

Add a high-pressure pre-alarm on PT-420 between the PIC-420 HI alarm (130 kPa g) and PSHH-426 (250 kPa g), with a written response: restore the vapour path (HV-422), call the header owner.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-18

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-18
Node	N5 · Vapour line to the header	Action party	Process
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.2	Flow MORE : More vapour to the header	- PCV-420 fails open (instrument air loss, positioner) - PIC-420 set point too low - HV-423 opened - feed over-heated or lighter	Drum depressures toward the header at several times the design vapour rate (see dynamic check): header pressure rises and upsets other users; liquid auto-refrigerates; heavier vapour.	- FI-420 HI alarm 900 Nm³/h - PIC-420 LO alarm 70 kPa g - TI-410 LO alarm - header pressure control (off-plot)	S3 L3 9 Medium
N7.8	Depressuring MORE : Depressuring too fast to the header	PCV-420 opened wide or HV-423 opened at shutdown	Auto-refrigeration of the drum liquid; header overload; liquid carry-over.	- FI-420 HI alarm 900 Nm³/h - TI-410 LO alarm	S3 L3 9 Medium

HAZOP recommendation

Check the fuel-gas header (and flare) capacity for PCV-420 failed open and for drum depressuring; consider a travel stop on PCV-420.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-19

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-19
Node	N4 · D-400 flash drum	Action party	Instrument / Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.1	Level MORE : High level	• LCV-400 fails closed • HV-411 gate drops • XV-418 closes spuriously • LT-400 freezes or reads low • receiver pressure high • feed above design • hot oil lost (less flash, more liquid)	Liquid reaches the mesh pad and vapour nozzle: liquid slugs into the vapour line and fuel-gas header (burner flooding and fire at users); vapour space lost, pressure rises.	• LIC-400 HI alarm 80 % • LG-400 • LSHH-416 (own sensor) > 90 % for 5 s → I-1 closes XV-408 and XV-409	S4 L3 12 High
N4.10	Measurement OTHER THAN : LT-400 reads wrong	• Transmitter freezes • impulse line plugged • turbulence or foam at the taps	LIC-400 moves LCV-400 on a false level: the true level can reach the vapour nozzle or empty unseen.	• LG-400 gauge glass (rounds) • LSHH-416 and LSL-417 have their own sensors	S4 L3 12 High
N4.11	Measurement OTHER THAN : PT-420 reads wrong	Transmitter drift (up to 40 kPa), impulse line fault	PIC-420 holds the wrong true pressure: products drift off spec; a transmitter reading low hides a real over-pressure.	• PSHH-426 own sensor • TI-410 against the bubble point	S3 L3 9 Medium
N6.1	Flow NO : No liquid outflow	• LCV-400 fails closed (positioner or air failure) • HV-411 gate drops • XV-418 closes spuriously • receiver isolated	D-400 level rises (N4 level MORE) to the high-high switch within minutes (see dynamic check); downstream unit starved.	• FIC-410 LO alarm 8 m³/h • LIC-400 HI alarm 80 % • LSHH-416 I-1	S4 L3 12 High

HAZOP recommendation

Add an independent level transmitter LT-400B with its own high-level alarm, and comparison alarms LT-400 vs the LSHH-416/LSL-417 level sensors and PT-420 vs PSHH-426; read LG-400 on every round.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-20

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-20
Node	N4/N6 · D-400 flash drum / Liquid line to the downstream unit	Action party	Process / Instrument
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.1	Level MORE : High level	- LCV-400 fails closed - HV-411 gate drops - XV-418 closes spuriously - LT-400 freezes or reads low - receiver pressure high - feed above design - hot oil lost (less flash, more liquid)	Liquid reaches the mesh pad and vapour nozzle: liquid slugs into the vapour line and fuel-gas header (burner flooding and fire at users); vapour space lost, pressure rises.	- LIC-400 HI alarm 80 % - LG-400 - LSHH-416 (own sensor) > 90 % for 5 s → I-1 closes XV-408 and XV-409	S4 L3 12 High
N4.2	Level LESS : Low level (liquid seal lost)	- LCV-400 fails open - HV-413 bypass left open - LIC-400 in MAN at a high output - LT-400 reads high - HV-415 drain left open	Vapour blows through LCV-400 into the downstream receiver built for liquid at 40 kPa g: receiver over-pressure, rupture, release; downstream pumps lose suction.	- LIC-400 LO alarm 20 % - LSLL-417 (own sensor) < 5 % for 5 s → I-2 closes XV-418 - LCV-400 fail closed	S5 L2 10 High
N6.2	Flow MORE : More liquid than made; gas blow-by	- LCV-400 fails open - HV-413 bypass open - LIC-400 in MAN at a high output - LT-400 reads high	The drum drains; once the level is lost, vapour at drum pressure blows through into the receiver built for liquid at 40 kPa g: over-pressure, release.	- FIC-410 HI alarm 18 m³/h - LIC-400 LO alarm - LSLL-417 I-2 closes XV-418 - receiver relief (off-plot)	S5 L2 10 High
N7.5	Timing LATE : Late response to the I-2 closure	- XV-418 closed by LSLL-417 while the feed keeps running - alarm not acted on	Level climbs from low-low to the high-high switch: I-1 trips the unit (see dynamic check).	- LIC-400 HI alarm 80 % - LSHH-416 I-1	S3 L3 9 Medium

HAZOP recommendation

Review the I-2 action: when XV-418 closes, alarm and cut the feed (or require it by procedure) so the level does not climb to LSHH-416; take the gas blow-by scenario to LOPA.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-21

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-21
Node	N6 · Liquid line to the downstream unit	Action party	Process
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.2	Level LESS : Low level (liquid seal lost)	- LCV-400 fails open - HV-413 bypass left open - LIC-400 in MAN at a high output - LT-400 reads high - HV-415 drain left open	Vapour blows through LCV-400 into the downstream receiver built for liquid at 40 kPa g: receiver over-pressure, rupture, release; downstream pumps lose suction.	- LIC-400 LO alarm 20 % - LSLL-417 (own sensor) < 5 % for 5 s → I-2 closes XV-418 - LCV-400 fail closed	S5 L2 10 High
N6.2	Flow MORE : More liquid than made; gas blow-by	- LCV-400 fails open - HV-413 bypass open - LIC-400 in MAN at a high output - LT-400 reads high	The drum drains; once the level is lost, vapour at drum pressure blows through into the receiver built for liquid at 40 kPa g: over-pressure, release.	- FIC-410 HI alarm 18 m³/h - LIC-400 LO alarm - LSLL-417 I-2 closes XV-418 - receiver relief (off-plot)	S5 L2 10 High

HAZOP recommendation

Confirm the downstream receiver relief for gas blow-by through LCV-400 and HV-413 fully open at the highest drum pressure, and that the LSLL-417 delay (5 s) is short enough.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-22

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-22
Node	N4/N5 · D-400 flash drum / Vapour line to the header	Action party	Mechanical / Process
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.6	Composition AS WELL AS : Liquid carried over with the vapour	• Foaming (surfactant in the feed after a tank switch) • level high • vapour rate above design (FCV-400 open, over-heated feed)	Liquid C6 in the vapour product: AI-420 below spec; liquid in the fuel-gas header, flame-out or flooding at the users' burners.	• AI-420 LO alarm 75 wt% • LIC-400 HI alarm • FI-420 HI alarm • LSHH-416	S3 L3 9 Medium
N4.7	Separation PART OF : Mesh pad removes only part of the droplets	Mesh pad damaged, displaced or plugged after an upset	Fine C6 mist in the fuel gas; condensate in the header low points; AI-420 slightly low.	• AI-420 LO alarm • header knock-out (off-plot)	S2 L3 6 Medium
N5.5	Composition LESS : Vapour below spec (AI-420 < 75 wt%)	• Feed heavier • feed over-heated • drum pressure low • carry-over	Off-spec fuel gas sold at a lower price; heavier gas condenses in the header (liquid in the fuel-gas system).	AI-420 LO alarm 75 wt%	S2 L4 8 Medium
N5.6	Temperature LESS : Vapour line cooled (condensation)	• Rain or cold ambient on an uninsulated line • heavy vapour	Condensate collects in low points and at PCV-420; liquid slugs reach the header.	• Line routing (to confirm) • none on the P&ID	S2 L3 6 Medium

HAZOP recommendation

Carry-over: inspect the mesh pad at each turnaround, provide an antifoam injection point, slope the vapour line to the drum and confirm the header knock-out drum and drains.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-23

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-23
Node	N1/N6 · Feed supply and flow control / Liquid line to the downstream unit	Action party	Operations / Instrument
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.8	Composition MORE : Feed lighter (more C4)	Tank switch to a lighter feed (butane toward 45 wt%)	More vapour and higher drum pressure demand on PCV-420; liquid product C4 above the 28 wt% spec sent downstream.	- AI-410 HI alarm 28 wt% - FI-420 HI alarm 900 Nm³/h - PIC-420	S2 L4 8 Medium
N1.9	Composition LESS : Feed heavier (less C4)	Tank switch to a heavier feed (butane toward 25 wt%)	Less vapour, more liquid; vapour carries more C6 so AI-420 falls toward its 75 wt% spec; LCV-400 opens further.	- AI-420 LO alarm 75 wt% - FI-420 LO alarm 300 Nm³/h - LIC-400	S2 L4 8 Medium
N4.6	Composition AS WELL AS : Liquid carried over with the vapour	- Foaming (surfactant in the feed after a tank switch) - level high - vapour rate above design (FCV-400 open, over-heated feed)	Liquid C6 in the vapour product: AI-420 below spec; liquid in the fuel-gas header, flame-out or flooding at the users' burners.	- AI-420 LO alarm 75 wt% - LIC-400 HI alarm - FI-420 HI alarm - LSHH-416	S3 L3 9 Medium
N5.5	Composition LESS : Vapour below spec (AI-420 < 75 wt%)	- Feed heavier - feed over-heated - drum pressure low - carry-over	Off-spec fuel gas sold at a lower price; heavier gas condenses in the header (liquid in the fuel-gas system).	AI-420 LO alarm 75 wt%	S2 L4 8 Medium
N6.5	Composition MORE : Liquid product too light (AI-410 > 28 wt%)	- Feed lighter - hot feed too cold - drum pressure too high	Off-spec liquid to the downstream unit (lower price); light ends flash in the receiver and raise its pressure.	- AI-410 HI alarm 28 wt% - TI-410 LO alarm 40 °C	S2 L4 8 Medium
N6.7	Measurement OTHER THAN : AI-410 frozen or wrong	- Analyzer sample line blocked - analyzer fault	Off-spec liquid not seen; the operator optimises on a false reading.	- Lab samples - TI-410 as an inferential check	S2 L3 6 Medium

HAZOP recommendation

Sample the feed at every tank switch and the products each shift; add a flat-line (frozen signal) alarm on AI-410 and AI-420.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-24

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-24
Node	N1 · Feed supply and flow control	Action party	Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.10	Composition AS WELL AS : Free water in the feed	Water carried from tank bottoms after a tank switch	Water boils in E-400 and D-400 (pressure surges), collects in the drum bottom and goes to the downstream unit; ice or hydrate in the let-down valve at low temperature; corrosion.	• None specific • HV-415 drain	S2 L3 6 Medium

HAZOP recommendation

Draw water from the feed tank before a switch and check the feed for free water; include a D-400 low-point water check in the rounds.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-25

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-25
Node	N4/N7 · D-400 flash drum / Procedures	Action party	Process / Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.6	Temperature LESS : Very cold liquid	Drum depressured quickly (PCV-420 fails open): auto-refrigeration	Liquid colder than normal in the bottoms line (the model gives about -0.5°C at the lowest, far above a typical -29°C minimum design metal temperature): ice and condensation on the line, valves stiff, off-spec liquid; brittle fracture is not credible on this unit.	TI-410 LO alarm 40°C	S3 L2 6 Medium
N7.7	Temperature (drain) LESS : Auto-refrigeration while draining	- Liquid drained under pressure through HV-415 to the closed drain - C4 flashes across the valve	Valve and drain line chilled toward the boiling point of butane (about -0.5°C at atmospheric pressure): ice on the valve can jam HV-415 open and release C4; cold burns. The model does not reach the carbon-steel minimum design metal temperature (typical -29°C).	- HV-415 normally closed - closed drain - procedure	S4 L3 12 High
N7.8	Depressuring MORE : Depressuring too fast to the header	PCV-420 opened wide or HV-423 opened at shutdown	Auto-refrigeration of the drum liquid; header overload; liquid carry-over.	- FI-420 HI alarm $900\text{ Nm}^3/\text{h}$ - TI-410 LO alarm	S3 L3 9 Medium

HAZOP recommendation

Write the drain and depressuring procedure against auto-refrigeration: slow, attended, double valves on HV-415, watch TI-410; confirm the minimum design metal temperature of D-400 bottoms, HV-415 and the drain line.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-26

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-26
Node	N7 · Procedures	Action party	Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.9	Composition OTHER THAN : Oxygen inside the drum	- Incomplete N2 sweep at start-up - air drawn in under vacuum - equipment opened and not purged	Flammable mixture inside D-400: internal explosion on ignition (static, pyrophoric scale).	- Start-up N2 sweep procedure - AI-400 (measures N2, not O2) - N2 blanket	S5 L2 10 High
N7.1	Purge (N2 sweep) LESS : N2 sweep incomplete	- Sweep time cut short - vented from one nozzle only (dead legs: LG-400, instrument nozzles, bottoms line) - AI-400 read before it settles	Air left in D-400 when hydrocarbon enters: flammable mixture, internal explosion on ignition.	- Procedure hold point - AI-400 (N2 only)	S5 L2 10 High

HAZOP recommendation

Make an oxygen check at the N2 sweep vent (portable meter) a hold point before hydrocarbon is admitted; sweep all dead legs (LG-400, instrument nozzles, bottoms line).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-27

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-27
Node	N7 · Procedures	Action party	Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.2	Sequence BEFORE : Hot oil before feed flow	• Step order not followed • TCV-401 or XV-409 left open from the last run	Stagnant feed in the E-400 tubes heated: vaporisation and blocked-in over-pressure (N2 pressure MORE).	Procedure: TIC-401 in MAN at 0 % and XV-409 closed until feed flows	S4 L2 8 Medium
N7.3	Sequence AFTER : Hot oil stopped after the feed at shutdown	• Wrong order • XV-409 left open	As above: blocked-in feed heated in E-400.	• Procedure • TIC-401 moves TCV-401 to bypass on a high TT-401	S4 L2 8 Medium
N7.4	Timing EARLY : Liquid sent downstream too early	• XV-418 opened or LIC-400 put in AUTO before the level is established • product sent before AI-410 is on spec	Gas blow-by to the receiver at low level; off-spec liquid downstream.	• XV-418 permissive: level above 10 % (SIS) • LSSL-417 I-2	S3 L2 6 Medium
N7.9	Maintenance OTHER THAN : Isolation incomplete	• Gate valves passing (HV-401, HV-411, HV-421) • XV-408/409/418 passing • no blinds • drain or vent not proven • hot oil not isolated	Hydrocarbon or hot oil released when equipment is opened: fire, burns, exposure.	• Permit to work • lock-out	S5 L2 10 High

HAZOP recommendation

Positive isolation for maintenance (blinds or double block and bleed at the feed, hot-oil, vapour and liquid battery limits), lock-out, depressure and N2 purge before opening; review the start-up and shutdown procedures against this study.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-28

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-28
Node	N3/N7 · Hot-oil side of E-400 / Procedures	Action party	Instrument / Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.7	Safety system OTHER THAN : XV-409 fails to close on demand	Valve stuck, solenoid fault, not proof-tested	On an I-1 trip hot oil keeps heating the blocked-in feed; the pressure rise continues.	• XV-408 closes the feed • TIC-401 moves TCV-401 to bypass • PSV-400	S4 L2 8 Medium
N7.10	Safety system OTHER THAN : SIS bypassed or reset with the cause present	• Bypass left after testing • forced reset	High level or high pressure unprotected.	• Reset permissives (level below 85 %, pressure below 230 kPa g) • XV-418 permissive above 10 %	S5 L2 10 High

HAZOP recommendation

SIS management: proof tests of LSHH-416, LSL-417, PSHH-426 and XV-408/409/418 at the interval set by LOPA; bypass permit with time limit; bypass status shown on the DCS.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-29

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-29
Node	N5 · Vapour line to the header	Action party	Safety
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.5	Temperature MORE : External fire at D-400	Pool fire from a leak at a flange, drain or valve station below the drum	Liquid boils off, the unwetted shell above the liquid overheats and fails below the PSV set pressure; large fire or vessel rupture.	- PSV-400 set 450 kPa g (fire case to be verified) - PSHH-426 I-1 stops feed and hot oil - closed drain	S5 L2 10 High
N5.7	Integrity OTHER THAN : Leak at the PCV-420 station	- Gasket or packing failure - vibration from choked flow across PCV-420	Flammable gas released at about 102.7 kPa g: flash fire or jet fire in the unit.	- None on the P&ID (no gas detection) - field rounds	S4 L2 8 Medium

HAZOP recommendation

Provide gas detection at the D-400 valve stations (to be set in the HAZID follow-up).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-30

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-30
Node	N7 · Procedures	Action party	Operations
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.1	Flow NO : No vapour to the header	- HV-421 gate drops (fails closed) - PCV-420 sticks closed - header isolated by others - HV-422 and HV-423 closed	Drum pressure rises at the flash rate (N4 pressure MORE): I-1 trip or PSV-400 lift to flare; liquid richer in C4 (AI-410 off spec) meanwhile.	- PIC-420 HI alarm 130 kPa g - FI-420 LO alarm 300 Nm³/h - PSHH-426 I-1 at 250 kPa g - PSV-400	S4 L3 12 High
N6.1	Flow NO : No liquid outflow	- LCV-400 fails closed (positioner or air failure) - HV-411 gate drops - XV-418 closes spuriously - receiver isolated	D-400 level rises (N4 level MORE) to the high-high switch within minutes (see dynamic check); downstream unit starved.	- FIC-410 LO alarm 8 m³/h - LIC-400 HI alarm 80 % - LSHH-416 I-1	S4 L3 12 High
N7.5	Timing LATE : Late response to the I-2 closure	- XV-418 closed by LSL-417 while the feed keeps running - alarm not acted on	Level climbs from low-low to the high-high switch: I-1 trips the unit (see dynamic check).	- LIC-400 HI alarm 80 % - LSHH-416 I-1	S3 L3 9 Medium

HAZOP recommendation

Train operators on the simulator on the scenarios of the dynamic check (loss of the vapour path, LCV-400 failures, TT-401 fault, hot-oil loss).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R-31

Project	Flash Drum Ver 1 · D-400 Operator Training Simulator	Document	FD-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R-31
Node	N1/N4 · Feed supply and flow control / D-400 flash drum	Action party	Instrument
Drawings	FD-400-PID-001 (piping and instruments) and FD-400-PID-002 (control and SIS)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.11	Measurement OTHER THAN : FT-400 reads wrong	- Orifice plate damaged - impulse line leak or plugged - wrong range after calibration	FIC-400 holds the wrong flow; the true feed rate is unknown; balances do not close and faults are misread.	- FI-410 + FI-420 as a rough balance - LIC-400 trend	S2 L3 6 Medium
N4.12	Relief OTHER THAN : PSV-400 passing	- Seat leakage after a lift - set-point drift	Continuous product loss to flare (cost, flaring); the valve may not reseal after the next lift.	- PSV tail-pipe temperature (field) - material balance	S2 L3 6 Medium

HAZOP recommendation

Add a DCS material balance (FT-400 against FI-410 + FI-420) with a deviation alarm, to reveal meter errors and a passing PSV-400.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			